

The Language Barrier Hiding Mathematical Competence

The Issue

Mathematics is commonly perceived as a challenging discipline, frequently eliciting discomfort and anxiety among learners. Recent statistics indicate that approximately 64% of Americans experience some level of math anxiety, with 13% reporting severe anxiety (Prodigy Education). These data highlight a pervasive struggle with mathematical concepts that may have substantial and lasting consequences.

Math anxiety often impedes academic and professional advancement, with nearly half of affected individuals reporting negative outcomes. (Eidlin-Levy et al.) It shapes career trajectories, restricts opportunities for advancement, and diminishes self-confidence. Notably, 37% of individuals report that math anxiety influences their financial decisions, including budgeting and investing. (Andersen et al.) These findings underscore the urgent need to address math anxiety to support both personal and professional development.

Math anxiety is prevalent among students but does not inherently reflect a deficiency in mathematical ability. (Caviola et al.) Research by Sheila Tobias (1978) and others demonstrates that challenges in advanced mathematics often stem from linguistic and psychological barriers rather than insufficient conceptual understanding. These barriers may include test anxiety, difficulties with technical language, or reluctance to seek assistance.

Students may experience challenges interpreting mathematical terminology or become overwhelmed by performance pressures, resulting in a cycle of anxiety that impedes academic progress. ("Math Performance and Academic Anxiety Forms, from Sociodemographic to Cognitive Aspects: a Meta-analysis on 906,311 Participants") Studies by Ma (525), Zhang et al., and Ferreira et al. emphasize the importance of addressing non-academic factors to foster a more inclusive and supportive educational environment.

Recognizing these barriers allows educators to implement targeted support strategies, such as providing clearer explanations, fostering a supportive classroom environment, and enhancing students' confidence in communicating mathematical ideas. Concrete classroom practices operationalize these strategies. For example, sentence frames like "To solve this problem, first I..." or "The key idea here is..." offer students accessible structures for articulating mathematical reasoning. Incorporating think-aloud protocols, in which teachers or students verbalize their problem-solving processes step by step, can demystify mathematical thinking. Peer explanation routines, where students explain their solutions or reasoning to a partner or small group, encourage active engagement and reinforce understanding. Empirical research supports the effectiveness of these interventions: a meta-analysis by Sammallahiti et al. (2023) found that structured cognitive and communicative strategies, such as sentence frames and peer explanation, significantly reduce math anxiety and improve student performance. Similarly, Ramirez et al. (2018) demonstrated that collaborative and inquiry-based classroom practices, including think-alouds, promote deeper understanding and lower anxiety levels among mathematics learners. Employing these strategies increases access to advanced mathematics and helps students overcome anxiety, thereby promoting greater academic achievement.

For advanced learners, including graduate students, early-career researchers, and professional mathematicians, these interventions can be adapted to address challenges at higher levels of expertise. Collaborative problem-solving groups that encourage open discussion of approaches can be integrated into research seminars or working groups, facilitating peer-to-peer explanation at a more advanced conceptual level. Think-aloud techniques may be employed when developing new proofs or exploring unfamiliar mathematical domains, making the reasoning process explicit and supporting metacognitive awareness. Structured mentoring relationships, in which experienced colleagues help reframing technical challenges or clarify advanced notation, can also reduce anxiety when engaging with new areas. Extending these strategies to advanced

contexts emphasizes clear communication, supports resilience when navigating complex research problems, and promotes ongoing professional growth.

Many individuals possess an intuitive grasp of mathematical concepts but encounter challenges with Greek letters, complex notation, and specialized jargon. (Verschaffel et al.) Everyday activities such as doubling recipes, coordinating time zones, planning routes, or estimating whether furniture will fit through a doorway require mathematical reasoning. These tasks depend on proportional reasoning and spatial geometry, which are frequently applied instinctively without formal terminology. (Banerjee and Bhat)

For example, although the term “diffeomorphism” may appear intimidating, its underlying concept is accessible: it refers to a smooth transformation that preserves an object’s properties, similar to stretching pizza dough. This analogy illustrates that advanced mathematical ideas are present in everyday contexts, making them comprehensible to a broad range of learners. Relating complex concepts to familiar experiences bridges the gap between intuition and formal mathematics, thereby enhancing engagement and understanding. (“What Makes an Analogy Effective for Students’ Learning of Mathematics? Conceptualizing Analogy Effectiveness for Teaching Mathematics”)

Many students report feeling overwhelmed by mathematics; however, these challenges are widespread and do not necessarily indicate a lack of ability. Confusion and anxiety frequently arise from difficulties with mathematical language. With appropriate support and effective strategies, students can overcome these obstacles. (“Math Performance and Academic Anxiety Forms, from Sociodemographic to Cognitive Aspects: a Meta-analysis on 906,311 Participants”)

Maintaining a mathematics vocabulary journal is an effective strategy. Students can record new terms, define them in their own words, and provide examples, thereby facilitating memorization and deepening understanding. Seeking explanations in everyday language or real-life scenarios further enhances comprehension. (Lin et al. 123-135)

Introducing formal mathematical concepts using familiar, accessible language helps students demystify the subject. (Gonzales-Martinez and Sanchez) Research indicates that intimidating language, rather than the content itself, often contributes to difficulties in mathematics (Tobias 65; Ferreira et al.). With appropriate instructional tools and methods, mathematics can become accessible and engaging for all learners.

Significance of the Issue

Financial Consequences of Avoidance

Math anxiety increases financial vulnerability in retail settings and contributes to wealth disparities. (Storozuk & Maloney, 2023) Those affected often experience negative emotions when faced with numerical information, which can cause cognitive overload and emotional distress. (Cohen & Korem, 2021) This discomfort may undermine their confidence and self-efficacy in handling numerical tasks, leading to avoidance. (Choe et al., 2019) As a result, they are less likely to compare prices, analyze interest rates, or evaluate investment options. (Storozuk et al., 2023) Over time, this avoidance leads to poor financial decisions and reduced opportunities for wealth accumulation (Peters, 2020).

Thirty-seven percent of respondents report that math anxiety impairs their financial decision-making, including budgeting, investing, and comparing loan terms. This statistic is drawn from a 2022 online survey of approximately 2,000 adults in the United States, conducted by Prodigy Education, which included a nationally representative sample balanced by age, gender, and socioeconomic background. Among Generation Z, 46% say math anxiety affects their financial decisions, such as budgeting, understanding bills, investing, and helping children with math homework (Prodigy Education).

Adults with high math anxiety are more likely to make costly financial errors, such as misunderstanding interest rates, overlooking key details in credit card offers, and avoiding financial products that require numerical evaluation (Lusardi and Tufano 340-345; 337; Ashcraft and Krause 245). Similar patterns have been documented in research from other countries, indicating that the impact of math anxiety on financial decision-making extends beyond the United States. (Lau et al., 2022) For example, studies in the United Kingdom, Australia, and Germany have reported comparable associations between math anxiety and mistakes in budgeting, borrowing, and investing (Furnham and Lumb 199; OECD 2019). These international findings reinforce the broader relevance of addressing math anxiety as part of financial education efforts.

Adults with math anxiety are less likely to compare mortgage rates, seek optimal financial products, understand compound interest, or make informed decisions about retirement and loans. This avoidance leads to overindebtedness and reduced wealth accumulation over time and is driven by anxiety rather than a lack of ability (Lusardi and Tufano 350-355). Research indicates that targeted educational interventions, such as math-specific cognitive behavioral strategies, numeracy workshops, and supportive learning environments, can help reduce math anxiety and improve financial decision-making skills. Integrating practical, real-life financial scenarios into math education has also been shown to increase confidence and support better financial outcomes (Ashcraft and Krause 250; OECD 2019).

Educational and Professional Consequences

Approximately 20% of Americans report missing significant career opportunities as a result of math anxiety, according to a national survey by Prodigy Education. The impact is particularly pronounced among Generation Z, with 62% of respondents indicating educational or career barriers related to math skills and 26% reporting lost job opportunities due to insufficient proficiency. While not all sources are peer-reviewed, organizational reports and large-scale surveys provide valuable insights into current trends. These findings highlight the substantial effect of math anxiety on the prospects of young professionals. (The Professional and Technical Workforce: By the Numbers, 2024)

Math anxiety is a global phenomenon; however, rates in the United States frequently exceed those observed in other developed countries. An OECD report indicates that 30% of American students feel nervous or tense when engaging in mathematics, compared to 12% in Finland, 14% in Japan, and 15% in Canada. These statistics demonstrate that American students experience math-related stress and avoidance at significantly higher rates than their international peers. Countries with lower rates of math anxiety often implement early interventions and supportive teaching methods, offering potential models for policymakers in the United States. (Coddington et al., 2023, p. 101229)

Math anxiety frequently impedes both academic and professional advancement. According to Prodigy Education, 47% of adults with math anxiety have experienced significant setbacks in education or career progression. Weir (year) further observes that many individuals trace the onset of math anxiety to adolescence, with 34% identifying middle school as the origin of their fears (Prodigy Education, Weir, Ahmed 158-166).

Mathematics serves as a foundational discipline for advanced studies in STEM fields, equipping students with essential skills for higher-level coursework and future academic or career objectives. However, many students experiencing math anxiety begin to avoid these courses before high school. (Eidlin-Levy et al.)

Such avoidance restricts immediate academic options and produces long-term consequences for the STEM talent pipeline. (Ferdinand et al., 2024) A reduced number of students pursuing these fields may diminish the pool of skilled professionals required for innovation in critical sectors (Paechter et al., 2024). Although math anxiety represents a significant barrier, additional factors such as socioeconomic status, access to high-quality resources, and extracurricular support also influence participation in advanced mathematics and STEM disciplines. (Lau et al., 2022) These trends underscore the urgent need for

targeted interventions and support systems to address math anxiety and foster engagement in mathematics and related fields. (Ferdinand et al., 2024)

Effective policy measures include investing in teacher professional development to address math anxiety, implementing curriculum reforms that prioritize conceptual understanding and student engagement, and providing early screening and support for at-risk students. Policymakers may also collaborate with universities and STEM employers to offer mentorship and real-world learning opportunities, thereby clarifying mathematical concepts and their practical applications. Collectively, these strategies can foster a supportive learning environment that builds confidence and promotes long-term academic achievement. (Niu et al., 2022)

To begin moving from theory to action, schools or districts can take several practical steps. First, school leaders might initiate a needs assessment to identify specific areas where students or teachers experience the greatest challenges with math anxiety. Based on these insights, they could organize professional development workshops focused on strategies for reducing anxiety, such as using clear language and collaborative learning activities. Schools can also pilot revised lesson plans or curriculum units that emphasize accessible explanations and relate mathematics to real-life contexts before expanding changes school-wide. Establishing early identification protocols—such as brief student surveys or teacher checklists—can help pinpoint students who would benefit from targeted support or small group interventions. Creating partnerships with local universities or businesses to establish mentorship programs can further enhance these efforts. By starting with manageable pilot initiatives and collecting feedback from students and educators, schools can refine their approaches and gradually scale up successful strategies to benefit the entire learning community.

Research identifies several primary self-reported triggers of math anxiety: fear of failure (53%), inadequate understanding (48%), calculation difficulties (46%), and mismatched teaching methods (45%). Notably, three of these factors pertain to instructional approaches rather than the subject matter itself (SWNS). Additional contributing issues include technical jargon, unexplained abstract symbols, rapid instructional pacing, and teaching practices that emphasize memorization over comprehension. (Lau et al., 2022) Evidence-based reforms such as inquiry-based learning, collaborative classroom activities, and real-world problem-solving have demonstrated effectiveness in reducing math anxiety. For instance, a meta-analysis by Ramirez et al. (2018) found that collaborative and inquiry-based instruction significantly reduced students' reported math anxiety. Similarly, Passolunghi et al. (2021) reported that real-world problem-solving tasks increased student confidence and decreased anxiety. These strategies foster deeper understanding and a more supportive learning environment, thereby addressing the root causes of math anxiety and enhancing educational outcomes.

Impact of Math Anxiety on STEM Fields

In 2024, individuals employed in STEM (Science, Technology, Engineering, and Mathematics) roles earned a median annual wage of \$103,580, which is more than double the \$48,000 median for non-STEM positions. Furthermore, STEM jobs are projected to grow by 8.1% from 2024 to 2034, nearly three times the 2.7% growth expected for non-STEM occupations. (Numbers, research, and discovery: STEM employment projected to take off!, 2025)

Math anxiety remains a significant barrier in STEM, hindering individual growth and deterring many qualified candidates from pursuing high-paying roles. Its effects extend beyond personal challenges, posing a critical economic concern that limits workforce diversity and overall economic growth (Bureau of Labor Statistics). Evidence-based strategies, including fostering a growth mindset, establishing peer support, and applying targeted classroom techniques, have proven effective in reducing math anxiety. (Employment in STEM occupations, 2024)

The United States may face up to 2 million unfilled STEM positions, primarily due to a significant skills gap among potential workers. (Employment in STEM occupations: U.S. Bureau of Labor Statistics, 2024) Research indicates that math anxiety is a

major contributor, often predicting STEM avoidance more strongly than math proficiency or past performance. (Daker et al., 2021)

A study by Ferdinand and colleagues shows that math anxiety independently affects students' STEM career choices, regardless of their math skills or achievements. The research, based on a nationally representative sample of over 1,500 high school students from diverse backgrounds, used surveys and academic records to analyze the relationships among math anxiety, performance, and career intentions. These results highlight the urgent need for targeted interventions and support systems to reduce math anxiety and build a more inclusive, skilled workforce.

Students with math anxiety often avoid mathematics courses during key academic years, resulting in the loss of essential prerequisites for many academic and career paths, as noted by Finlayson. This avoidance limits educational options and creates a cycle where reduced exposure to math increases anxiety and widens the skills gap. The ongoing cycle of avoidance and anxiety further restricts opportunities, posing a significant barrier for aspiring STEM students. (Rozgonjuk et al., 2020)

Consequently, many students interested in STEM view these fields as out of reach, contributing to workforce shortages. Only about 20% of graduating U.S. high school students are prepared for college-level STEM coursework, highlighting a pipeline issue related to early math avoidance (iD Tech). (Gagnaniello, n.d.) Targeted interventions are needed to address math anxiety and promote early engagement with mathematics.

The Impact of Anxiety on Career Choices in STEM Fields

A recent study by Ferdinand, Malanchini, and Rimfeld (2024) finds a strong link between math anxiety and the avoidance of certain career paths among emerging adults, especially in science, technology, engineering, and mathematics (STEM). The research shows that individuals with high math anxiety are much less likely to pursue STEM careers, regardless of their actual math skills. These results suggest that emotional and psychological barriers have a greater impact on career choices than mathematical ability. One strength of the study is its focus on emerging adults, offering insight into a critical stage of career decision-making.

However, the study relies on self-reported measures of anxiety and career intentions, which may introduce bias or limit the generalizability of its findings. As a result, many qualified individuals are discouraged from entering STEM fields, which reduces both diversity and innovation in the STEM workforce. These findings suggest that targeted policy actions are needed, such as increased funding for interventions that address math anxiety and the integration of psychological support within educational standards. By using this research to inform policy, decision-makers can support initiatives that help more students overcome anxiety barriers and pursue careers in STEM.

Expanding on this perspective, Özdemir (2023) examines how math anxiety influences career interests and decisions. The study highlights that math anxiety can significantly lower self-efficacy, or the belief in one's ability to succeed. This reduction in self-efficacy occurs as individuals who experience anxiety about mathematical tasks often engage in avoidance behaviors, such as steering clear of math-related subjects or situations. Additionally, they may engage in negative self-talk, reinforcing beliefs that they are incapable of succeeding in math. Anxiety about mathematical tasks often reduces self-confidence and causes individuals to doubt their skills and potential.

As a result, individuals with lower self-efficacy may disengage from academic and career paths that require math skills. The study shows that reduced self-efficacy creates a negative feedback loop: as confidence drops, interest in math-related fields declines as well. This disengagement increases anxiety and perpetuates a difficult cycle of avoidance. However, the cycle can be disrupted by interventions such as providing early positive math experiences, encouraging mentorship programs, or creating supportive classroom environments that help students build confidence and resilience in math. These strategies can foster self-efficacy and offer practical pathways to break the pattern of avoidance.

Özdemir's findings show that math anxiety has a significant and complex impact. It is not just a minor obstacle to learning math but a major barrier that can undermine self-confidence and discourage individuals from pursuing STEM careers. If left unaddressed, these psychological barriers may continue to limit the participation of qualified individuals in these fields.

Gender and Equity in Mathematics: Structural and Social Determinants

Hottinger's analysis demonstrates that gender, race, and societal perceptions shape the definition of a "mathematician." Mathematical identity is constructed through narratives that have historically marginalized women and people of color in mathematics (Hottinger). These narratives influence both self-perception and external perceptions, reinforcing stereotypes that discourage participation. In school settings, such narratives can manifest in classroom interactions, teacher expectations, and the representation of students in advanced coursework.

Studies indicate that about 70% of women experience math anxiety, compared to 57% of men (Prodigy Education). This anxiety disproportionately undermines women's academic performance and self-perception in mathematics (Yu et al.). Stereotypes portraying mathematics as a male-dominated field foster feelings of inadequacy among women, discouraging participation regardless of ability. High-achieving girls are less likely than boys to pursue math-intensive careers, primarily due to heightened anxiety rather than a lack of ability (Denervaud et al.). Math anxiety has broader consequences, discouraging students from pursuing STEM careers and diminishing self-confidence. This issue is particularly acute among women, who report higher anxiety levels throughout their educational trajectories, from elementary school through higher education (Özdemir 7). Research indicates that gender disparities in math performance can emerge early, sometimes before formal schooling, even when preschoolers of different genders demonstrate similar mathematical skills. (Devlin et al., 2024)

Statistical evidence indicates that 70% of women experience math anxiety, compared to 57% of men (Prodigy Education). This anxiety more severely affects women, diminishing their academic performance and self-image in mathematics (Yu et al.). Stereotypes depicting mathematics as male-dominated contribute to feelings of inadequacy among women, discouraging engagement regardless of skill. High-achieving girls are less likely than boys to pursue math-intensive careers, primarily due to increased anxiety rather than a lack of ability (Denervaud et al.).

Math anxiety discourages students from pursuing STEM careers and undermines self-efficacy. This issue is particularly pronounced among women, who report higher anxiety at every academic stage, from elementary school through higher education (Özdemir 7). Research demonstrates that gender disparities in math achievement emerge early, often before formal schooling, even though preschool children of different genders exhibit similar abilities. (Penner & Paret, 2008)

Math anxiety influences gendered career choices, with high-performing girls more likely to avoid math-intensive paths due to emotional responses rather than ability (Denervaud et al.). Math anxiety also negatively affects women's test performance, classroom participation, and confidence, even when their achievements are comparable to those of boys. This highlights the need for targeted interventions to help students, particularly girls, overcome psychological barriers (Opesemowo et al., 2025). Persistent self-doubt leads many young women to avoid advanced mathematics coursework, limiting both immediate academic opportunities and future potential in STEM fields (Samuel et al., 2022, pp. 613-626).

The intersection of math anxiety with race, ethnicity, and socioeconomic status further complicates the challenges students encounter. Women of color and those from low-income backgrounds encounter additional barriers, including stereotype threat, limited access to quality resources, and underrepresentation in advanced mathematics. Racial and ethnic minority students experience heightened math anxiety due to persistent stereotypes, resource disparities, and a lack of role models in advanced math classes (Steele 613; Ma 530). The absence of role models and limited access to effective instruction intensify anxiety and discourage participation. (Maloney et al., 2013, pp. 115-128) Negative stereotypes diminish students' sense of belonging, as societal biases erode confidence and impede mathematical success ("Math-Failure Associations, Attentional

Biases, and Avoidance Bias: The Relationship with Math Anxiety and Behavior in Adolescents” 1001-1011). Collectively, these factors increase anxiety and have enduring effects on engagement with mathematics.

Peer interactions significantly shape students’ self-perceptions and mathematical abilities, influencing their engagement with mathematical concepts. These interactions can enhance or undermine self-esteem and motivation, which is particularly important in a challenging subject. (Reciprocal relations between adolescents’ self-concepts of ability and achievement emotions in mathematics and literacy, 2021) Media portrayals of gender roles in mathematics reinforce stereotypes and shape students’ perceptions of their potential. For example, repeated depictions of men as math experts may lead girls to internalize beliefs of lesser capability. (Wille et al., 2018) The combined impact of these influences shapes academic choices and self-concept, highlighting the need to address disparities within the educational system (The Secret Language of Peers: How Peer Behaviors Signal Mindset and Influence Classroom Experiences, 2024). This evidence suggests that performance gaps are primarily driven by environmental and cultural factors rather than inherent ability.

External factors exert a greater influence on students’ performance than presumed innate abilities (Denervaud et al.). This finding underscores the need for systemic change in educational and societal structures to promote equity in participation and achievement in mathematics.

Teachers, peers, and media play essential roles in shaping students’ experiences and sense of belonging in mathematics. Teachers who set high expectations and provide encouragement can foster positive attitudes toward mathematics. (Wu & Cai, 2023, pp. 1234-1245) Addressing these factors is critical for creating a more inclusive and equitable STEM education environment.

Physical, Psychological, and Physiological Harm

The phrase “I am not a math person” has become a significant identity label shaped by societal norms, gender expectations, and educational experiences (Weir). For example, research by Good, Rattan, and Dweck (2012) found that girls are less likely to be encouraged to pursue advanced mathematics, and studies of classroom interactions show teachers tend to call on boys more often for challenging math questions (Sadker & Sadker, 2009). Girls may also see fewer female mathematicians featured in textbooks or hear teachers and parents suggest that boys are naturally better at math. Media representations often reinforce these biases, showing men as engineers or scientists and women in non-mathematical roles. (Nandi et al., 2024) A large-scale study conducted by Ceci and Williams (2010) revealed that such societal cues and expectations significantly impact girls’ participation and performance in mathematics. This mindset is internalized and influences how individuals perceive their mathematical abilities. (Zhang et al., 2019)

These social and cultural influences are not only important for educational practice; they also raise foundational questions in mathematical epistemology and philosophy. The question of whether mathematical ability is socially constructed or innate draws us directly into longstanding philosophical debates, notably the positions of Platonism, constructivism, and social constructivism. Platonism, for example, asserts that mathematical truths exist independently of human minds, are perceivable by reason alone, and are universal in nature. However, if our beliefs about who can do mathematics are socially constructed and shaped by experience, this standpoint comes into question, inviting a move toward constructivist and social constructivist perspectives. Constructivism holds that mathematical knowledge is actively built by each learner through experience, while social constructivism emphasizes the fundamental role of language, culture, and interpersonal interaction in the formation of mathematical concepts. Such considerations invite us to reflect on the nature and development of mathematical knowledge, the role of social context in shaping reasoning, and the epistemic access individuals have to mathematics itself. Furthermore, the idea that access to mathematics is mediated by cultural context invites us to question the presumed objectivity and universality of mathematical knowledge, as social constructivism would argue that what we take as mathematical truth is, to a significant extent, created and negotiated within communities. If the identities and experiences of

learners affect not only who engages with mathematics but also how mathematical truths are approached and understood, then the discipline itself cannot be fully disentangled from its social foundations. This perspective suggests a more dynamic and situated view of mathematical knowledge, where objectivity is informed by the plurality of human experiences and the evolving practices of mathematical communities. By explicitly connecting these social influences to major philosophical positions, it becomes clear that mathematics is not a purely abstract and neutral discipline, but one that evolves in dialogue with culture, language, and society. (Rieggle-Crumb & Humphries, 2012)

Hersh and John-Steiner examine the emotional aspects of mathematics, such as love, hate, anxiety, and confidence. They argue these emotions are learned and shaped by social environments and instructional quality (Hersh and John-Steiner). Supportive classrooms foster positive attitudes, while negative experiences can lead to anxiety and aversion. The impact of emotion extends beyond attitudes, directly influencing mathematical reasoning and creativity. For example, research on cognitive load theory suggests that anxiety can consume working memory resources, which limits a student's ability to solve complex problems. Conversely, feelings of curiosity or enjoyment may increase intrinsic motivation and encourage flexible thinking, leading to creative approaches and deeper conceptual understanding. (Moustafa et al., 2020, pp. 287-296) Emotional states can act as either barriers or catalysts: a student who feels confident is more likely to persevere with challenging problems, while one who feels threatened may rely on rote memorization or abandon reasoning altogether. Theories such as Fredrickson's broaden-and-build model suggest that positive emotions help broaden cognitive resources, supporting better problem-solving and creative insight in mathematical contexts. (Fredrickson, 2004)

Schwartz examines whether mathematical ability is innate or learned and concludes it develops through experience, instruction, and practice. Empirical research supports this view: for example, Siegler and Ramani (2008) found that structured number board games can significantly improve numerical proficiency in young children from low-income backgrounds, while the longitudinal studies of Gunderson et al. (2012) demonstrated that parental involvement and positive math talk at home lead to measurable gains in children's achievement. Additionally, a meta-analysis by Sisk et al. (2018) showed that growth mindset interventions, which emphasize effort and learning, can lead to statistically significant improvements in students' math performance. These studies indicate that educational challenges typically result from poor teaching, limited resources, or language barriers, rather than fixed cognitive limitations. Educators are encouraged to improve instructional methods and learning environments to support all students. (Heyd-Metzuyanim, 2025)

Math anxiety produces measurable physiological effects, such as increased cortisol, elevated heart rate, and impaired working memory, similar to the body's response to physical threats. This often leads to avoidance, creating skill gaps that reinforce fear and further avoidance, known as the "anxiety-performance cycle" (Ashcraft and Krause). Their research highlights psychological barriers to mathematical development.

Gold and Simons point out that even professional mathematicians discuss what counts as a valid argument. For instance, some proofs rely on visual or probabilistic reasoning, which may spark debate among experts about their rigor or acceptability. A well-known example is the use of computer-assisted proofs, such as the proof of the Four Color Theorem, which initially faced skepticism because the argument relied on exhaustive case checking rather than traditional, concise written logic. Similarly, disputes can arise over the validity of non-constructive existence proofs versus constructive proofs, with mathematicians disagreeing about whether a proof adequately demonstrates an object's existence when it does not provide an explicit example. The negotiation of these standards reflects the evolving nature of mathematical rigor. As mathematical methods and technologies advance, the community actively debates and revises what qualifies as an acceptable proof. These debates are deeply connected to foundational questions in the philosophy of mathematics, including whether mathematical truth is discovered (as in Platonism), constructed (as in constructivism), or derived from formal systems of symbols without reference to external reality (as in formalism). The controversies surrounding acceptable proof methods often reflect underlying philosophical commitments about the nature and existence of mathematical objects, the role of intuition, and the necessity of explicit construction versus abstract existence. As such, these evolving standards of rigor not

only shape mathematical practice but also directly engage with epistemological issues concerning what constitutes mathematical knowledge and truth. These evolving standards influence professional practice by requiring mathematicians to adapt their communication, teaching, and collaboration to ensure clarity, consensus, and transparency. These cases illustrate how mathematical communication is often more complex and context-dependent than conventional teaching suggests, and highlight the importance of presenting reasoning in line with professional standards.

Numeracy, in a mathematically precise sense, refers to the ability to understand, interpret, and use mathematical concepts, operations, and reasoning in varied contexts. Unlike general quantitative literacy, which involves a broader set of skills such as interpreting graphs or using numerical information in daily life, numeracy encompasses the proficiency to engage with the logical structure of mathematics, apply quantitative methods, and communicate mathematical ideas with rigor. This distinction is important for mathematicians, as numeracy involves not just the manipulation of numbers, but also familiarity with mathematical argumentation, abstraction, and problem-solving strategies. (Schreiber-Barsch et al., 2020)

Numeracy is crucial for overall well-being, extending beyond just academic achievement. For students, numeracy affects daily decisions, such as managing a budget, understanding interest rates on student loans, accurately interpreting data found in news articles or social media, and making informed choices about nutrition and health. Higher levels of numeracy are linked to better health, increased wealth, and enhanced decision-making skills. Individuals with low numeracy skills are more susceptible to cognitive biases, which can lead to poorer choices and contribute to a 'hidden tax' in a data-driven society. Additionally, low health numeracy results in less effective medical decisions and higher healthcare costs. (Rolison et al. 222-234, Root & Bhala, 2020) As Peters (2020) points out, improving numeracy is essential for both individual and public health. Peters' (2020) review found that stronger numeracy skills lead to greater competency in assessing risk and benefit, improved health outcomes, and reduced susceptibility to misleading information. Peters also recommends implementing targeted educational policies and interventions to support numeracy development from an early age, arguing that these steps are critical to empowering individuals and ensuring equitable access to information, services, and opportunities.

Numeracy also forms the foundation for advanced mathematical practice and is vital for professional mathematicians and researchers. The ability to reason quantitatively, interpret abstract symbols, and apply logical structures is essential not only in daily life but also in the rigorous development of mathematical theory. Advanced research depends on strong numeracy skills for constructing proofs, understanding complex models, and communicating sophisticated ideas with clarity and precision. Thus, strengthening numeracy benefits individuals at every level, from students to expert practitioners, reinforcing the discipline and advancing mathematical knowledge as a whole. (Amalric and Dehaene 4909-4917, Gal et al., 2020)

Research by Lyons and Beilock utilized fMRI scans to examine the neurological reactions of individuals with high math anxiety. They found that when anticipating math-related tasks, the brain areas associated with physical pain were activated. This response was triggered by anxiety surrounding mathematical language and symbols, rather than the act of problem-solving itself (Lyons and Beilock). The results suggest that the body reacts to the mere presence of mathematics, rather than to the reasoning involved in mathematical processes.

Carey et al.'s 2019 review synthesized decades of research on math anxiety and identified three interrelated dimensions: cognitive, affective, and motivational.

1. **Cognitive Channel:** This dimension involves difficulties with working memory when handling mathematical notation. For those with math anxiety, complex symbols and abstract concepts can overwhelm cognitive resources, hindering problem-solving, reasoning, and the ability to understand relationships. Increased anxiety may cause mental blocks that limit access to necessary skills and knowledge. Individuals may struggle to recall formulas or apply principles correctly, leading to greater inadequacy and frustration. This decline in performance undermines confidence and perpetuates a negative cycle, significantly limiting mathematical ability and success in academic and practical contexts.

2. **Affective Channel:** This dimension addresses the emotional reactions people experience when facing mathematical problems. For many, even thinking about mathematics can cause panic, frustration, and inadequacy, increasing stress and associating math with fear. As distress grows, individuals often avoid math by missing classes, avoiding exams, or abandoning math-related careers, which reinforces their fears. Over time, this avoidance heightens feelings of inadequacy and negative perceptions, with lasting effects on academic performance, confidence, and self-esteem. Mathematics thus becomes both a cognitive and emotional challenge.
3. **Motivational Channel:** This dimension involves avoiding subjects, careers, and activities that require mathematical skills. Individuals with math anxiety often withdraw from opportunities that could enhance their abilities and boost their confidence. This reluctance shapes their educational and career choices, leading them to steer clear of mathematics and limit their professional growth. By avoiding math-related experiences, they miss out on valuable skill development and continue to experience anxiety, creating a self-fulfilling prophecy of incompetence. These patterns hinder both personal and professional development, resulting in lower confidence and competence in mathematics. The factors involved are interdependent and create a self-reinforcing cycle that perpetuates math anxiety. Traditional interventions often fail to be effective because they focus solely on cognitive difficulties, overlooking the equally important emotional and motivational aspects shaped by how mathematical concepts are presented (Carey et al.). In response, advanced interventions for higher levels of mathematical training have begun to address these broader dimensions. For example, structured mentoring programs, peer-supported collaborative problem-solving, and workshops that explicitly address emotional barriers aim to build resilience and confidence among students and early-career mathematicians. Reflective teaching practices, such as integrating growth mindset strategies and fostering a supportive classroom community, have also shown promise in mitigating anxiety and sustaining motivation as learners progress to more advanced mathematical work. Such approaches are particularly relevant for educators and mentors seeking to support individuals beyond introductory courses.

According to Souviney's research, children learn mathematical reasoning from their everyday experiences, but they need direct instruction to master formal mathematical language. The difference between informal understanding and formal performance is known as a "translation gap," not a deficit, and children require support to overcome it (Souviney 218-222).

The Influence of Mathematical Notation on Working Memory

Lyons and Beilock's fMRI study, *When Math Hurts*, identified math anxiety as a neurocognitive phenomenon. The researchers recruited 28 adults who were pre-screened and categorized as either high or low in math anxiety based on standardized self-report measures. (Hart & Ganley, 2019, pp. 122-139) During the experiment, participants underwent brain imaging while solving a series of timed math puzzles, such as basic arithmetic and algebra problems, as well as word puzzles designed to serve as a non-mathematical control condition. The task order was counterbalanced, and both puzzle types were matched for difficulty and presentation format to isolate the neural effects of math-specific anxiety. (Ashcraft & Krause, 2007) This controlled design allowed the researchers to compare brain activity associated with mathematical versus linguistic tasks, and to assess how individual differences in math anxiety modulated anticipation and performance in these contexts. (Sarkar et al., 2014)

The primary finding was that anticipating engaging in mathematics, rather than the act itself, triggers discomfort. In individuals with high math anxiety, anticipating a math task activated the bilateral dorsoposterior insula, a brain region associated with physical pain (Lyons and Beilock e48076). This neural activation was absent during actual math performance, suggesting that the perceived threat is emotional and perceptual rather than computational. As the authors note, "For someone who has math anxiety, the anticipation of doing math prompts a similar brain reaction as when they experience pain" (Lyons and Beilock e48076). This perspective reframes math anxiety as an issue of emotional regulation in response to symbolic cues, rather than a deficit in reasoning. (González-Gómez et al., 2023) This interpretation is consistent with related findings that math anxiety is distinct from math competence (Lyons and Beilock, "Mathematics Anxiety" 2102).

This perceptual threat has a direct impact on working memory. Two complementary theoretical frameworks elucidate this mechanism:

1. **The Working Memory Depletion Model:** Ashcraft and Krause demonstrated that math performance depends on working memory for arithmetic tasks that extend beyond simple recall (243). (Ashcraft & Krause, 2007, pp. 243-248) Individuals with high math anxiety exhibited reduced working memory spans, particularly on computation-based tasks (Ashcraft and Krause 244). Anxiety consumes cognitive resources, disrupts attentional control, and impairs information updating, thereby diminishing processing efficiency in mathematical tasks (Ashcraft and Krause 246). (Ashcraft & Krause, 2007, pp. 243-248)

In practical terms, math anxiety fills working memory with intrusive, self-referential thoughts, thereby reducing the capacity available for problem-solving and vocabulary retrieval (Ashcraft and Kirk 224). (Shi & Liu, 2016) Ashcraft and Kirk observed that university students with high math anxiety performed worse on number-based working memory tasks than their non-anxious peers, a deficit attributed to reduced working memory capacity resulting from anxiety (226).

2. **Attentional Control Theory (ACT):** Eysenck et al. propose that anxiety impairs executive functions by depleting working memory resources (Eysenck et al. 968). Efficient cognition depends on the balance between goal-driven and stimulus-driven systems. Anxiety disrupts this equilibrium, allowing the stimulus-driven system to predominate (Eysenck et al. 986). (Eysenck et al., 2007, pp. 336-353) For learners with math anxiety, unfamiliar notation serves as a threat stimulus that captures attention and further reduces resources available for problem-solving. (Caviola et al., 2017)

Notation is central to this process. According to Cognitive Load Theory (CLT), developed by John Sweller, total cognitive load comprises intrinsic, extraneous, and germane components (Sweller, "Cognitive Load During Problem Solving" 258). Extraneous load refers to the cognitive effort imposed by poorly designed instructional methods that impede schema acquisition (Sweller et al. 11). Concrete examples from mathematics include the use of the same symbol for different concepts within a single context, such as the letter 'x' being used to represent both a variable and a multiplication operation. Similarly, ambiguous or overloaded notation, such as using f' to denote the derivative of a function while also using prime notation to indicate distinct but related objects, can confuse learners. Mathematical expressions that stack multiple indices, as in tensor notation (e.g., $T^{i_{jk}}$), often create additional difficulty for students who are not yet fluent in these symbolic conventions. (Hinrichs, 2018)

These notational choices can increase extraneous cognitive load, requiring students to devote attention to deciphering symbols rather than understanding underlying concepts. (Ashcraft & Krause, 2007, pp. 243-248) Empirical research supports these claims: Kirshner (1989) demonstrated that students confronted with unconventional or complex notational systems experienced measurable increases in problem-solving time and error frequency, attributable directly to working memory overload. Similarly, Durkin and Rittle-Johnson (2012) found that clearer and more intuitive notation in instructional materials led to improved learning outcomes and lower extraneous cognitive load, as measured by both performance and subjective cognitive load ratings. These studies quantify the impact of notational complexity, illustrating the need for educational researchers to address symbol design as an integral component of effective mathematics instruction.

Unfamiliar or complex notation is a major source of extraneous cognitive load. While it does not increase the complexity of the mathematical concept itself, it increases the effort required to complete the task. (Ayres, 2001, pp. 227-248) When notation differs from a learner's intuitive understanding, the effort to decode it can overshadow their ability to use conceptual knowledge (Rada and Lucietto 120-123). The impact of notation is especially pronounced across mathematical fields that rely on highly specialized symbolic languages. For example, topology uses visually dense diagrams and shorthand symbolic systems, while mathematical logic introduces multi-tiered symbols that often defy everyday experience. (Atagi et al., 2016) Notably, research on the cognitive impact of notation among expert mathematicians reveals that even those with extensive

experience are not immune to the effects of complex or unfamiliar symbolic systems. Studies have shown that experts also experience temporary increases in cognitive load and reduced problem-solving efficiency when required to switch to newly introduced or unconventional notation, underscoring the universal nature of these processing demands. (Vallée-Tourangeau et al., 2016) Even experienced mathematicians can encounter periods of adjustment when learning discipline-specific notational conventions, highlighting that notational complexity is not reserved for novices. (Nuerk et al., 2004, pp. 835-863)

Recent CLT research shows that adding representational systems requires extra processing and integration, increasing cognitive load. (Paas & Sweller, 2012) Working memory tasks that require high executive control, such as parsing dense notation, show a stronger negative correlation with math anxiety than simple storage tasks (Finell et al.). (Finell et al., 2022) Recent studies continue to highlight this issue: Lin and Kalyuga (2021) found that the introduction of overly symbolic or dense mathematics notation in instructional contexts elevated cognitive load ratings and reduced student performance, even among upper-level undergraduate learners. Similarly, a large-scale study by Nguyen, Rauterberg, and Hu (2023) demonstrated that aligning notation with students' prior knowledge substantially mitigates unnecessary processing demands and improves comprehension.

Furthermore, international comparative work by Hitt et al. (2019) underscores the need to consider cultural and linguistic variation in symbol conventions, as misalignments can unintentionally increase cognitive load for multilingual learners or those from diverse educational backgrounds. (Ngu et al., 2023, pp. 1051-1063). These findings reinforce the importance of continual evaluation and adaptation of notation to ensure accessibility and lower barriers for all learners.

The typical experience of a learner with math anxiety follows this sequence:

- **Cue:** The learner encounters complex or unfamiliar notation, which prompts an immediate sense of confusion.
- **Threat Response:** This confusion triggers anticipatory anxiety, activating the brain's pain and threat-detection networks, as demonstrated by Lyons and Beilock (e48076). The learner experiences increasing tension and anticipates potential failure or embarrassment.
- **Resource Depletion:** As anxiety intensifies, worry and intrusive thoughts consume cognitive resources, particularly the phonological loop and central executive functions of working memory (Ashcraft and Krause 244). This reduction in available capacity impairs the learner's ability to focus on the task.
- **Extraneous Load Spike:** Decoding unfamiliar notation imposes additional cognitive demands, further straining an already burdened working memory system (Sweller et al. 11). As a result, the learner finds it increasingly difficult to process information.
- **Performance Drop:** As cognitive capacity diminishes, the learner's performance declines significantly. This phenomenon, often described as "choking under pressure," illustrates how anxiety can directly impair mathematical ability and execution (Beilock 339). The outcome is frequently frustration and disappointment.

Interventions targeting this cycle provide empirical support for the model. Beilock's laboratory found that expressive writing about math anxieties before testing can reduce worry and free working memory resources, thereby improving performance. (Park et al., 2014, pp. 103-111) Instructional redesigns that employ consistent, visually distinct, and conceptually clear notation also enhance knowledge transfer without altering intrinsic content. (Atagi et al., 2016) For example, conceptually clear notation avoids overloading symbols with multiple meanings within the same context, and ensures that each symbol consistently represents a single concept. Visually distinct notation uses formatting such as spacing, color, or varied symbol styles to differentiate variables and operations clearly and to standardize conventions across materials, reducing confusion. (Moscoso et al., 2022) This finding aligns with Sweller's principle that "when part of the total cognitive load is used up by extraneous load, the learning is less than optimal". (Sweller, 2010)

However, the scope and duration of these interventions' effects may be limited. For example, while expressive writing can produce immediate benefits, its long-term impact on math anxiety and academic achievement remains under investigation, and follow-up data are sparse or confined to short timeframes. Similarly, instructional changes to notation often show the greatest benefit in controlled educational settings, which may not generalize to more varied or less supportive classroom environments. Effectiveness can also vary with individual differences in baseline anxiety, subject-matter complexity, intervention duration, and prior knowledge. Several studies note that small sample sizes, brief interventions, lack of longitudinal tracking, and limited participant diversity further constrain the robustness and generalizability of the findings. As a result, these interventions may not fully eliminate anxiety or extraneous cognitive load, but instead provide incremental improvements that can support learning, especially when combined with broader instructional strategies. Researchers and practitioners should pay particular attention to these limitations—especially those involving sample size, intervention duration, and transferability to real-world settings—when designing or interpreting intervention studies to ensure realistic expectations regarding practical significance. (Guo, Park et al., 2014)

In summary, mathematical notation is not a neutral element. For learners with math anxiety, it functions as both an emotional trigger and a cognitive burden, consuming the working memory resources necessary for successful performance. These findings suggest that efforts to reform or develop mathematical notation could have a meaningful impact on learning outcomes. By prioritizing notation that is more intuitive, consistent, and accessible, mathematicians and educators could reduce unnecessary cognitive load and help lower barriers to entry for a wider range of students. (Atagi et al., Scheibe et al., 2023)

However, the feasibility of large-scale notation reform is complex and brings significant challenges. Implementing changes to established mathematical notation would require broad consensus within the mathematical community, as consistency is essential for effective communication and the continuity of knowledge. Rapid or fragmented reforms risk creating confusion, disrupting the transmission of ideas across subfields, and potentially isolating generations of learners and researchers accustomed to existing systems. Philosophically, such reforms also raise questions about the nature of mathematical language and whether mathematical meaning is inherently tied to specific symbols or is independent of notation. Achieving successful reform would therefore require careful negotiation, pilot testing, and collaboration among educators, researchers, and institutions to ensure both increased accessibility and the preservation of shared standards necessary for the discipline's progress and cohesion.

The implications of notation reform also extend beyond teaching and learning, impacting mathematical research practice itself. More accessible, standardized notation can facilitate clearer communication among researchers from different subfields or institutions, thus supporting interdisciplinary collaboration. Minimizing ambiguity and reducing the cognitive demands of deciphering unfamiliar symbolic systems may enhance the efficiency of peer review and publication processes. Additionally, clear and inclusive notation can make advanced mathematical work more approachable to new entrants to the field, broadening participation and encouraging cross-pollination of ideas. Such considerations invite ongoing evaluation of symbolic systems to ensure that advances in mathematical thought are matched by clarity and inclusivity in their representation. ("Mathematics education researchers' practices in interdisciplinary collaborations: Embracing different ways of knowing" 1-19, Specht & Crowston, 2022)

Several strategies could guide concrete steps for notation reform. One approach is to establish interdisciplinary committees composed of mathematicians, educators, cognitive scientists, and linguists, tasked with reviewing and recommending standardized symbol conventions for new and existing areas of mathematics. Developing and maintaining open-access repositories or databases of standardized notation would facilitate adoption and reference. In addition, creating formal guidelines for symbol selection, prioritizing clarity, consistency, and avoiding overload, can help minimize unnecessary complexity. Piloting new or revised notational systems in diverse educational settings, followed by empirical assessment of cognitive impact and learning outcomes, would ensure that revisions genuinely reduce extraneous cognitive load. Further,

incorporating user-centered design practices, such as soliciting feedback from students and international scholars, can help ensure the accessibility of notation for learners from varied backgrounds.

At the same time, it is important to acknowledge that notation reform faces significant challenges. Resistance may arise from longstanding mathematical traditions, the deep integration of current symbolic systems into research and teaching, and concerns about disrupting communication across subfields or generations. Widespread changes to mathematical notation risk fragmenting the field and creating transition issues for both new and established practitioners. These complexities highlight the need for careful consensus-building and thoughtful implementation when considering large-scale reform, balancing the benefits of accessibility with the necessity of maintaining coherence and clear communication within the mathematical community.

While these systemic reforms are important, individual educators can also take immediate steps to make notation more accessible in their own classrooms. Teachers might clarify the meaning and purpose of each new symbol as it is introduced, use consistent notation throughout lessons, and provide students with symbol glossaries or reference sheets. Where possible, instructors can adapt notation to align with students' existing knowledge, explicitly explain ambiguous or overloaded symbols, and encourage students to verbalize their interpretations of symbols during problem-solving activities. Adding color, spacing, or visual cues to distinguish similar symbols can further reduce confusion. Encouraging students to ask questions whenever notation feels unclear and inviting feedback on which symbols are most challenging can help foster a more inclusive environment. By making these targeted adjustments, individual educators can immediately reduce cognitive barriers and support understanding, even before broader notation reforms are implemented. These strategies are equally relevant and adaptable to higher education contexts, such as undergraduate and graduate mathematics courses, where instructors can clarify advanced notation and encourage students to discuss and interpret symbols in research seminars or advanced problem sets. By tailoring these practices to the needs of students at all levels, educators can help make even the most complex mathematical material accessible. These strategies provide a framework for ongoing reform and research, leading to practical improvements in both education and professional mathematical communication. (Atagi et al., Greiner-Petter et al., 2020)

Math Anxiety as a Linguistic Rather Than Conceptual Barrier

Millions of people who claim to be “bad at math” successfully navigate topological problems daily—tying shoes, braiding hair, wrapping gifts, untangling jewelry, securing loads with rope, knitting, crocheting, and countless other activities that require sophisticated topological competence (Matsumoto and Grishanov 120–22). The disconnect between this demonstrated competence and mathematical self-concept exemplifies how linguistic barriers to formal mathematics obscure genuine mathematical ability operating in embodied, spatial, and tactile domains (Sossinsky 210–15). (Robertson and Graven)

Parents negotiating bedtime with children, shoppers timing purchases for sales, employees deciding how hard to work when monitoring is imperfect—all demonstrate an intuitive understanding of strategic equilibria, dominant strategies, and repeated-game dynamics (Raeburn and Zollman, Rubinstein 130–40).

People navigate topological concepts constantly—realizing that untangling headphones requires understanding how the cord is knotted—all without invoking Euler characteristics or homeomorphisms. A person untangling jewelry chains demonstrates sophisticated topological reasoning (knot theory): they visualize how loops connect, identify where chains are genuinely knotted versus merely tangled, and perform specific manipulations to reduce topological complexity. This is expert-level topology applied in real-time, lacking only the formal vocabulary to describe what's being done. Mathematical competence exists; the technical language that would allow them to communicate with professional topologists does not. (Croom and Firestone)

Everyone engages in stochastic reasoning daily, like deciding to bring a jacket in case of rain or leaving early for a flight due to potential traffic. Diversifying investments is another example of this. While formal mathematics, such as Markov chains, allows for precise predictions, the core idea that “the future is uncertain and should be modeled probabilistically” is widely understood. People make sophisticated decisions about uncertainty without needing specific mathematical terms, recognizing that while things may be predictable on average, they remain uncertain on an individual level. Understanding randomness does not require technical vocabulary; people can grasp complex ideas without formal language. (“Reasoned Decision Making Without Math? Adaptability and Robustness in Response to Surprise” 869-883)

Every person who plays a 3D video game, uses a smartphone’s orientation sensor, or watches a CGI movie demonstrates fluency with quaternion-based transformations through their intuitive understanding of 3D rotation. They can predict how objects will rotate, judge whether orientations are correct, and manipulate 3D spaces with precision. Yet if shown the equation $q = \cos(\theta/2) + \sin(\theta/2)(xi + yj + zk)$ and asked to explain quaternion rotation, most would claim they “don’t understand the math.” The mathematics is operating correctly in their interactions; only the symbolic representation is unfamiliar. This perfectly illustrates the document’s central thesis: mathematical competence exists independently of mathematical language. (Díaz et al.)

The competence-language gap is particularly stark in number theory. A child who can determine “If today is Tuesday, what day will it be 100 days from now?” is performing modular arithmetic: $100 \equiv 2 \pmod{7}$, so it will be Thursday (Dudley 3). The child may solve this by counting by sevens (“7, 14, 21... 98 is 14 weeks, so 2 days after Tuesday”) without ever encountering the formal notation. The mathematical reasoning is complete and correct; only the symbolic language is absent. This pattern recurs throughout number theory: people demonstrate numerical competence—recognizing patterns, applying divisibility shortcuts, estimating prime factorizations—without the formal vocabulary to discuss these concepts with mathematicians. (Göbel et al. 17-25)

Every user of online banking, encrypted messaging, or secure purchases trusts number-theoretic principles, understanding that encryption keeps their data safe. They can recognize the need for security and choose appropriate tools without knowing specific mathematical concepts like Euler’s totient function or modular arithmetic (Petrus 700–05). As Lefton observes, teaching cryptography is most effective when familiar problems are introduced before complex notation (Lefton 54–60). This supports the idea that mathematical literacy doesn’t require fluency in mathematical language. (White)

Every person understands genus, a topological invariant that influences how objects are used (Weinstein). For example, a bowl ($g = 0$) can’t be reshaped into a mug ($g = 1$) without creating a hole. A typical pair of eyeglasses (without lenses) has $g = 2$, while a button with four holes has $g = 4$. A wedding ring has $g = 1$, and two interlocked wedding rings form a more complex structure with genus 2 but remain distinct from a double-torus, as they can be separated by continuous deformation. This illustrates the intuitive understanding of linked versus fused rings—a fundamental topological insight. (Auckly 330-342)

To make the connections between advanced mathematical concepts and everyday experience clearer, consider these concrete examples:

- When tying shoes, braiding hair, or untangling jewelry, individuals are intuitively applying principles from topology and knot theory, such as understanding how loops connect and can be manipulated.
- Doubling a recipe, converting measurement units in cooking, or planning routes involves proportional reasoning and spatial geometry.
- Deciding to bring an umbrella based on weather forecasts, or leaving early for a flight to avoid traffic, demonstrates probabilistic thinking, similar to concepts in statistics and stochastic processes.
- Managing a budget, comparing shopping prices, or understanding a credit card’s interest rate requires arithmetic, percentage calculations, and numeracy skills.

- Playing 3D video games or using a phone's orientation sensor relies on an intuitive sense of spatial rotations, which is closely related to quaternion mathematics used in computer graphics.
- Children solving questions like, "If today is Tuesday, what day will it be 100 days from now?" are performing modular arithmetic without realizing it.

These examples show that advanced mathematical ideas are present and practiced continuously in daily life, often without the formal vocabulary or notation. By recognizing and relating these familiar contexts to abstract mathematical concepts, educators can help bridge the gap between intuition and formal mathematics, enhancing understanding and reducing intimidation. (Miyejav)

Jargon and Notation as Mechanisms of Gatekeeping in Mathematics

Ortlieb et al. argue that each academic discipline, including mathematics, forms a distinct discourse community defined by specialized vocabulary, syntax, and argumentation. These elements are not intuitive and must be explicitly taught as part of disciplinary literacy. (Planas and Pimm) In mathematics, this literacy involves fluency in reading, writing, and communicating according to established conventions, which differ from general computational skills and require understanding mathematical expression in written form. This focus on disciplinary literacy aligns with foundational debates in the philosophy of mathematics and epistemology, particularly regarding the nature of mathematical knowledge and the relationship among language, representation, and understanding. (Herheim) By highlighting the constructed and communicative aspects of mathematical language, scholars show that mathematics is not merely a set of facts or procedures, but a practice grounded in human conventions and discursive norms. (Planas and Pimm)

Disciplinary literacy and communicative conventions play a critical role in advanced mathematics. Mastery of mathematical language and adherence to established norms are essential for effective research collaboration and the clear exchange of complex ideas. Mathematical publications require not only rigorous proofs but also precise exposition, with specific expectations for notation, terminology, and structure. These conventions shape the formulation of theorems, promote clarity, and reduce ambiguity. (Peterson et al. 1-10) As a result, disciplinary literacy both facilitates understanding and defines the standards of advanced mathematical research and communication.

Hiebert further developed this foundation by examining how learners acquire proficiency with mathematical symbols. He showed that symbolic fluency, which is essential for engaging with mathematical content, can be taught and learned independently of conceptual understanding. This distinction indicates that students may have strong reasoning skills but still struggle with symbolic representation. The separation between reasoning and symbolic fluency supports Hiebert's language-barrier thesis, which argues that mastery of mathematical language is crucial for student success.

Ferreira et al. (2025) examined the link between mathematics-specific vocabulary and early mathematical performance in young students. They found that proficiency in mathematics-specific vocabulary is a stronger predictor of mathematical performance than general language skills. Students who struggle with mathematical terminology often experience increased anxiety and stress, which negatively impact their academic outcomes. As a result, complex mathematical jargon can obscure students' true abilities and lead to outcomes that do not accurately reflect their potential.

A systematic review by Castillo et al. (2025) assessed mathematics teaching methods and found significant drawbacks in relying solely on formal mathematical notation, which can hinder student learning. The review stressed that without concrete, intuitive representations, students may struggle to grasp key concepts. To promote deeper understanding, multiple semiotic representations—including symbols, graphs, and algebraic expressions—should be integrated with real-world contexts. However, many curricula still prioritize formalism over intuitive understanding, which may limit students' ability to comprehend mathematical principles fully. (Højgaard) While much of this research focuses on early and secondary education, the effects of notation and representation also impact higher levels.

In advanced mathematical research, the growing abstraction and complexity of formal notation can create barriers for those unfamiliar with specialized conventions. For example, in algebraic geometry, notation for concepts like sheaf cohomology or schemes can make published work inaccessible to mathematicians outside the field, sometimes hindering interdisciplinary collaboration. For instance, the notation $H^i(X, F)$ denotes the i th sheaf cohomology group of a sheaf F on a scheme X , where understanding the role of each symbol requires background knowledge of both topological and algebraic structures. Similarly, the notation $\text{Spec}(R)$ is used to refer to the spectrum of a ring R , which has specific meaning in the context of schemes but may not be transparent to those outside algebraic geometry. These specialized notational systems can present significant obstacles to comprehension, even for mathematicians from closely related areas.

In contrast, the use of universal notation in group theory, such as standard symbols for group operations and homomorphisms, has improved communication and supported cross-disciplinary research. Inconsistencies in terminology between subfields have historically been confusing; for example, the term “ring” in algebra versus analysis led to misunderstandings until shared conventions were adopted.

Even among experienced mathematicians, clear exposition and varied representations are essential for understanding and collaboration. This shows that challenges with notation and representation affect all levels of mathematical practice. (Hernandez-Martinez et al.)

The American Psychological Association (APA) published a 2023 report indicating that math anxiety is primarily an emotional response rather than a consequence of cognitive deficiencies. Many individuals experiencing math anxiety possess adequate mathematical skills, yet anxiety inhibits their ability to demonstrate these skills, particularly in high-pressure contexts.

The report highlights the importance of environmental factors for individuals with math anxiety. In supportive, stress-reducing environments that encourage a positive approach to mathematics, these individuals often perform as well as peers without math anxiety. This finding shows that emotional states significantly impact performance, suggesting that the challenges faced by those with math anxiety are primarily emotional rather than numerical.

The research shows a statistically significant mediating effect of anxiety on mathematical performance. Anxiety is closely tied to personal identity, self-perception, and beliefs about mathematics. These results suggest that difficulties experienced by individuals with math anxiety are mainly psychological and linguistic, rather than due to a lack of understanding of mathematical concepts. (Živković et al.)

These insights call for a reevaluation of teaching strategies in mathematics education. The study recommends supportive approaches that build student confidence and reduce anxiety. Emotional well-being should be a core element of mathematics education, not secondary to skill development. The report advocates for reforms that emphasize mental health and emotional resilience alongside academic success (Özdemir 10).

Demedts et al. advanced the understanding of math anxiety by distinguishing between state and trait math anxiety. State math anxiety is an acute, situational panic during mathematical tasks, while trait math anxiety is a chronic, persistent fear of mathematics. Their findings show that trait math anxiety has a greater impact on performance, especially in complex tasks. This effect increases when tasks involve multiple forms of presentation, such as symbols, technical notation, and formal terminology. Therefore, the way information is presented can trigger anxiety in individuals with trait math anxiety, rather than reflecting a lack of mathematical understanding.

Kirschner examined the relationship between linguistic competence and mathematical ability, arguing that many mathematical challenges arise from linguistic processing difficulties rather than reasoning deficiencies. He emphasizes that

language proficiency and mathematical skills are distinct, so individuals may excel in one area while struggling in the other. This perspective highlights the complexity of math anxiety and its varied effects on learners.

Tobias argues that “math block” often results from language barriers rather than difficulties with mathematical concepts. Many students struggle with notation, vocabulary, and formal argumentation, not with mathematical logic itself. He asserts that supportive instruction focused on interpretation and translation can significantly reduce or eliminate mathematics anxiety (Tobias 65-70). Unfortunately, institutions often misinterpret these struggles as deficiencies in competence, leading to reliance on remedial courses and repetitive drills (Hodkowski). If language barriers are the root cause, traditional instructional approaches are inadequate. Instead, educational efforts should connect the complex language of mathematics with students’ intuitive reasoning abilities (Espinass and Fuchs).

Wynn and Reyes examine the rhetorical aspects of mathematical communication, arguing that mathematical arguments are persuasive acts embedded in language rather than simply conveying objective truths. They maintain that how mathematics is articulated and presented significantly affects access to understanding, making rhetorical skills essential for mathematical literacy (Wynn and Reyes). This perspective intersects with ongoing philosophical debates about the objectivity and truth of mathematics. While traditional positions like Platonism view mathematics as objective and discoverable, rhetorical and linguistic perspectives suggest that mathematical knowledge is shaped by human language and conventions. This raises questions about its universality and neutrality. Considering these philosophical issues highlights the complex relationship between mathematical communication and perceptions of objectivity within and beyond the discipline.

While specialized vocabulary and complex proof structures in formal mathematics can act as gatekeeping mechanisms, excluding those unfamiliar with disciplinary norms and contributing to feelings of inadequacy among students with linguistic challenges, it is important to recognize their positive aspects as well. (Leyva and Joseph 1187-1197) Formalism and precise terminology promote clarity, enable creative abstraction, and support rigorous reasoning, all of which are valued in the mathematical community. In advanced research and problem-solving, such rigor often leads to significant breakthroughs and ensures ideas are communicated with minimal ambiguity. At the heart of these practices are two influential philosophical perspectives on mathematical knowledge: essentialism and constructivism. Essentialist approaches view mathematical truths as existing independently of human experience, emphasizing universal, objective principles and the need for precise definitions and formal proof. Constructivist perspectives, by contrast, emphasize that mathematical knowledge is actively constructed by learners through experience, language, and cultural context, and that meaning emerges from individual or communal engagement with mathematical ideas.

Balancing these perspectives presents both risks and benefits. An overly essentialist focus can reinforce barriers by prioritizing abstraction and established formalism at the expense of accessibility, making mathematical discourse exclusive and less adaptable to diverse learners. On the other hand, a strictly constructivist approach, if not supported by rigorous standards, risks undermining the universality and transferability that allow mathematics to advance and link communities globally. For educators and the mathematical community, the central challenge is to find a productive middle ground: to uphold high standards and nurture creative abstraction while increasing accessibility and inclusivity. Many leading mathematicians now advocate for approaches that retain the rigor of proof and abstraction while presenting definitions, notation, and exposition as clearly and welcomingly as possible for all learners. Priority can be given to scaffolding learning so that students are gradually introduced to formalism, with support that connects intuitive reasoning to systematic structures. Maintaining high standards of rigor does not require sacrificing inclusion. Instead, transparency in definitions, thoughtful use of terminology, and clear, gradual progression to abstraction can allow a broader spectrum of students to participate fully in mathematical discourse while upholding the tradition of mathematical excellence.

The Importance of Framing in Mathematics Education

Pei, Poon, and Suen (2025) conducted a comprehensive study underscoring the critical role of mathematical engagement in mediating the relationship between anxiety levels and academic performance in mathematics. Their findings indicate that when students engage with mathematical concepts through accessible language and real-world examples, anxiety levels decrease significantly. This evidence suggests that the mode of presenting mathematical content may exert a greater influence than the content itself. Similarly, Finlayson demonstrated that introducing mathematical concepts in clear, everyday language before formal notation substantially enhances students' confidence and performance. This sequential approach, which first fosters understanding through familiar language and then transitions to formal mathematical expressions, significantly enriches the learning experience (Finlayson). These perspectives align with established theories such as Vygotsky's concept of the Zone of Proximal Development, which emphasizes the importance of scaffolding and the use of familiar contexts to support learning, as well as Bruner's theory of constructivist learning, where approachable language and real-world references help bridge new concepts. These findings can also be related to Piaget's theory of cognitive development, which suggests that students progress through stages of understanding and benefit from instruction that matches their cognitive readiness, and to Skemp's distinction between instrumental and relational understanding, indicating that relational approaches using accessible language can foster deeper comprehension. Additionally, Tall's framework on concept formation highlights the transition from embodied and symbolic representations to formal abstraction, reinforcing the value of first anchoring mathematical ideas in familiar contexts. Collectively, these findings support instructional strategies that prioritize accessibility and relatability, thereby improving student engagement and achievement.

Historically, perspectives on mathematical ability and pedagogy have shifted significantly. In the past, prevailing views often regarded mathematical talent as an innate trait, accessible only to a select few. This belief influenced traditional pedagogical approaches that emphasized rote memorization and procedural mastery, frequently marginalizing students who did not immediately excel. (Pool and Ericsson) More recent research, such as Byrnes's analysis—which synthesizes findings from multiple large-scale longitudinal studies involving student cohorts at various educational stages—provides substantial evidence that mathematical competence develops gradually through three core components: skill acquisition, conceptual growth, and linguistic fluency. Byrnes draws on data from studies with sample sizes ranging from several hundred to several thousand participants, using mixed-methods approaches that combine quantitative assessment of student performance with qualitative evaluation of learning strategies. This framework challenges the idea that some students inherently lack mathematical ability, suggesting instead that many have not been exposed to effective, accessible teaching methods tailored to their needs. As a result, their potential often remains unrealized until appropriate educational strategies are implemented (Byrnes). These findings underscore the importance of creating supportive and inclusive learning environments that promote all students' understanding and mastery of mathematical concepts and skills.

A comprehensive longitudinal study by Holenstein et al. examined transfer effects on the development of mathematical literacy, defined as the ability to interpret, communicate, and reason with mathematical ideas across contexts. The study found that language-based competence is a stronger predictor of transfer effects than procedural skills, underscoring the importance of linguistic fluency in promoting mathematical flexibility and adaptability (Holenstein et al. 810-820). In her advocacy for "Mathematical Competence for All," Abrantes identified significant institutional and pedagogical barriers to equitable mathematics education, which are mainly structural and linguistic rather than cognitive. Viewing mathematics as a universally accessible language, rather than an exclusive code, enables all learners to achieve mathematical competence (Abrantes).

A recent meta-analysis by Al-Naim and Mefi (2023) demonstrates the effectiveness of interventions to enhance language support in mathematics education. Their analysis found that structured approaches, including clear explanations of mathematical terminology and a gradual introduction to notation, consistently reduce mathematics anxiety among students. This anxiety reduction enables students to approach mathematical tasks with greater confidence and leads to measurable performance improvements. Practical language support interventions, such as maintaining vocabulary journals for key mathematical terms or using sentence frames to scaffold mathematical discussions, have shown particular promise in

facilitating comprehension and reducing student apprehension. These findings emphasize the importance of addressing both linguistic and pedagogical factors in mathematics instruction, as these elements are pivotal in shaping students' learning outcomes (Al-Naim and Mefi 875-880).

Hodkowski's analysis of current educational practices addresses the ongoing debate regarding the importance of conceptual versus procedural understanding in mathematics education. The findings indicate that cultivating a deep understanding of the reasoning behind mathematical concepts—the “why” of mathematics—fosters more robust and transferable mathematical thinking. Conversely, exclusive reliance on procedural fluency can leave students struggling with algorithms and terminology without a strong conceptual foundation. Even students who intuitively understand underlying concepts may encounter difficulties with procedural execution, highlighting the necessity for deeper engagement with mathematical material (Hodkowski).

Abbott et al. provide an in-depth analysis of the “math wars,” a term describing decades of debate over mathematics curriculum and pedagogy. Their research demonstrates that disputes over teaching methods are closely linked to broader societal issues, such as disparities in educational access and the pursuit of equity among diverse student groups. Abbott et al. argue that debates in mathematics education encompass not only pedagogical philosophies but also fundamental questions of educational justice and the legitimacy of varied mathematical knowledge. These complexities indicate that discussions in mathematics education extend beyond instructional strategies to address core issues of fairness and institutional responsibility (Abbott et al.). Similar debates and equity issues have been observed internationally, with countries such as the United States, the United Kingdom, and Australia reporting parallel controversies regarding curriculum design and access. While the specific local context and policy responses may differ, the underlying challenges surrounding fairness, access, and the legitimacy of diverse mathematical practices consistently emerge within educational systems worldwide. (Vithal et al. 153-164)

Research by Nunes, Schliemann, and Carraher provides strong evidence supporting ethnomathematics. Their findings indicate that integrating mathematical reasoning into meaningful, real-world contexts leads to higher accuracy and increased self-confidence among students. This effect is especially pronounced compared to traditional methods that emphasize abstract formal notation, which can leave students feeling disconnected from the material (Nunes et al.).

A study by Castillo et al. investigated the effects of gamified and active learning strategies on student engagement and performance in mathematics. These approaches aim to bridge the gap between informal reasoning, grounded in everyday experience, and formal mathematical notation. The results show that such strategies significantly reduce intimidation associated with complex mathematical concepts, thereby enhancing student motivation. This effect is particularly evident at the university level, where students often face the challenges of advanced mathematical notation (Castillo et al.).

A comprehensive meta-analysis by Sammallahti et al. (2023) synthesized 50 studies involving 9,125 participants. The analysis reported moderate effect sizes for interventions targeting math anxiety reduction ($g = -0.467$) and overall improvement in math performance ($g = 0.502$). The most effective interventions combined cognitive support, which fosters understanding and skill acquisition, with emotional regulation techniques that help manage anxiety and build resilience. Additionally, longer interventions, particularly those for students aged 12 and older, produced the most substantial effects. These findings suggest that integrating anxiety reduction with skill enhancement creates a more holistic and effective approach to mathematics learning (Sammallahti et al.).

For advanced or elite learners, these results highlight that even at high levels of achievement, addressing math anxiety and building emotional resilience remain pivotal. To offer more concrete strategies for this group, targeted mentorship programs can be implemented through pairing students with experienced mathematics mentors who guide advanced problem-solving, provide feedback on complex tasks, and support through academic challenges. Structured cognitive support interventions, such as individualized coaching sessions focusing on metacognitive strategies, collaborative research projects, and workshops

on tackling unfamiliar or competition-level problems, can also equip elite students with tools for sustained high performance. Incorporating peer-led seminars, opportunities for reflective journaling about problem-solving processes, and regular check-ins to discuss progress and setbacks further contributes to ongoing development. For example, a recent university-based mentorship program for advanced mathematics students matched upper-level undergraduates with graduate student mentors. Over the course of a semester, participants not only reported a measurable reduction in math anxiety as assessed by the MARS (Mathematics Anxiety Rating Scale), but also demonstrated greater willingness to tackle open-ended research tasks and present their findings in departmental seminars. Feedback from participants emphasized that regular mentor meetings and collaborative workshops helped them develop coping strategies for high-pressure situations, with several students choosing to enroll in more advanced mathematics courses the following term. These approaches help advanced learners build resilience, maintain motivation, and effectively navigate the pressures inherent to high-level mathematics. (Scheibe et al.)

The Ongoing Cycle of Math Anxiety

Intergenerational Transmission of Math Anxiety

Recent research demonstrates that math anxiety is transmitted across generations through both genetic and environmental factors. While Malanchini et al. (2022) is cited, it is important to clarify the methodological rigor of the principal studies. The primary empirical foundation is a landmark longitudinal field study published in *Psychological Science*, which used repeated measures over a school year to track changes in both parents and children. This design allowed the researchers to control for pre-existing math ability and other potential confounds, strengthening the validity of the claimed intergenerational effects.

Maloney, Ramirez, Gunderson, Levine, and Beilock (2015) tracked first- and second-grade students over a school year. The study found that children of highly math-anxious parents learned less mathematics and developed greater math anxiety, but only when those parents frequently assisted with homework (Maloney et al. 1480). The authors note: “Although there may be a genetic component to math anxiety, the fact that parents’ math anxiety negatively affected children only when they frequently helped on the child’s math homework” suggests an environmental transmission pathway (Maloney et al. 1480).

This finding has been replicated and extended:

1. **Genetic and Environmental Dual Pathway:** Quantitative genetic research, primarily relying on classic twin studies, indicates that approximately 40% of the variance in math anxiety is attributable to genetic factors, either directly or through their influence on mathematical ability and general anxiety (Wang et al., qtd. in Malanchini et al.). These studies compare monozygotic and dizygotic twins to estimate heritability, allowing researchers to disentangle genetic influences from shared environmental effects. Malanchini et al. further specify that 75% of this genetic variance is related to math attitudes, abilities, and achievement, highlighting substantial overlap between affective and cognitive traits (Malanchini et al.). Additionally, a study in *Frontiers in Psychology* confirmed that children’s math anxiety correlates with their arithmetic performance and is associated with parents’ educational level and attitudes (Malanchini et al.).
2. **Mechanisms of Environmental Transmission:**
 - **Reduced Positive Home Numeracy:** Parents with high math anxiety participate less frequently in positive numeracy activities, such as counting games, cooking measurements, budgeting, and number talk. When they do assist, they are more likely to use controlling rather than autonomy-supportive styles, which partially explains the association between parents’ math anxiety and children’s achievement (O’Sullivan et al.).
 - **Modeling Negative Attitudes:** Parents’ math anxiety is communicated to children through everyday activities, such as budgeting or calculating a tip, as well as through involvement with homework (Maloney et al. 1482). Longitudinal data indicate that children’s perception of their parents’ enjoyment of mathematics moderates the relationship between parental and child math anxiety, with the effect strongest in same-gender pairs, particularly mother-daughter pairs (Van Mier et al.; Girard and Russell).

- **Language Barrier Effect:** Parents who avoid math-related discussions limit their children's exposure to informal mathematical vocabulary. Research published in PMC [PubMed Central] demonstrates that math anxiety impairs symbolic, but not nonsymbolic, fraction skills, suggesting a perceptual basis. Children who lack early exposure to symbolic language encounter greater challenges when formal notation is introduced (Starr et al.).

An intervention published in Science tested this model directly. Berkowitz et al. (2015) provided math-anxious parents with a math-storybook app designed to encourage non-anxious interaction. Children whose parents used the app showed greater math growth, and the intergenerational link between parent and child anxiety was eliminated. However, it is important to note that the intervention sample was limited to families in the United States and focused on early elementary students, with a relatively modest sample size. Further, while the findings are promising, replication in larger, more diverse populations is needed to confirm the approach's effectiveness and generalizability. This demonstrates that transmission is malleable and not solely genetic.

Meta-Analytic Evidence: Consistent, Moderate, Negative Effects

The negative correlation between math anxiety and achievement is among the most consistently replicated findings in educational psychology.

Zhang, Zhao, and Kong (2019) conducted a meta-analysis [Frontiers in Psychology - PubMed] across 84 samples and 8,680 participants, reporting a significant negative effect size ($r = -0.32$), consistent with prior studies and reviews (Zhang et al.). This effect size reflects a moderate and robust relationship. To help contextualize, correlations of this magnitude are similar to the relationship between smoking and lung cancer, and are larger than the association typically found between self-esteem and academic achievement. (Barroso et al. 134-168) In practical terms, a correlation of $r = -0.32$ means that students with higher math anxiety are much more likely to perform lower in mathematics compared to their less anxious peers. (Barroso et al.) Moderator analysis revealed that the association was stronger for tasks involving innovative problem-solving than for rote recall, and more pronounced in senior high school than in elementary school, suggesting that the effects accumulate as the complexity of notation increases. (Namkung et al. 213-227)

This $r = -0.32$ estimate is consistent with findings from other large-scale syntheses:

- **Barroso et al. (2021) meta-analysis [PubMed - 131 studies, 478 effect sizes]:** Found $r = -.34$ for school-aged students (Barroso et al.). Moderation analyses indicated stronger effects for more advanced math and for students with lower working memory capacity.
- **Ma (1999):** The first major meta-analysis in the field, published in the Journal for Research in Mathematics Education, analyzed 26 studies and found $r = -.27$ between anxiety toward mathematics and achievement (Ma 520). Ma also found that the correlation was stronger for language-based tasks, such as word problems, than for pure calculation, reinforcing your language thesis. As later reviewers note: "In his 1999 meta-analysis, Ma compared the correlations of samples using the MARS ($k = 15$) to samples using other, non-MARS scales ($k = 22$)" and found consistent effects (Barroso et al.).
- **Hembree (1990):** The foundational meta-analysis of 151 studies found $r = -.34$, establishing the historical baseline.

Explanations for stronger effects in senior high school: Zhang et al. and Radišić et al. attribute the increase to both cognitive and semiotic factors. Radišić et al. (2015), analyzing PISA data from six European countries in the European Journal of Psychology of Education, found that the degree of early formality and abstraction significantly predicted anxiety at both the institutional and individual levels (Radišić et al. 10-15). Schools that prioritized conceptual understanding and delayed intensive use of formal notation reported lower student anxiety. (Muncer et al.)

This finding is consistent with recent evidence from ScienceDirect: “Accordingly, researchers have found correlations between various MA measures are significantly higher than those between MA and test anxiety. Similarly, a significant negative correlation has been found between MA, but not GA, and math performance” (Chang and Beilock 2016, qtd. in Orbach et al.). The association between math anxiety and performance is specific and mediated by working memory, rather than by general anxiety, as confirmed by a Frontiers meta-analysis on working memory as a mediator (Finell et al.).

Ding’s research on developmental students and recent MDPI studies on mathematical language self-efficacy indicate that reducing mathematics anxiety requires both internal resources, such as self-regulation, and explicit instruction in mathematical language with structured support during key tasks (Kopparla et al.). (“Self-Regulation and Mathematics Anxiety: The Conditional Mediating Role of Mathematical Language Self-Efficacy and Implications for Inclusive Education”) Notably, advances in cognitive psychology have shown that working memory plays a central role in mediating the relationship between anxiety and mathematical performance; anxiety consumes cognitive resources, leaving less capacity available for processing complex notation and multi-step problem solving. (Finell et al.) Further, language processing abilities are strongly linked to the understanding and use of mathematical symbols, since students must map natural language to formal notation and manipulate symbolic representations. (Landy and Goldstone) Neurocognitive studies suggest that difficulties in symbolic processing, such as translating word problems into algebraic expressions, are associated with heightened activation in brain regions related to both anxiety and language, underscoring the cognitive load imposed by complex notation. (Daroczy et al.) When formal notation is introduced without linguistic bridging- meaning without connecting intuitive vocabulary to symbolic forms- students with strong reasoning skills are systematically marginalized.

D’Ambrosio’s Ethnomathematics Framework

D’Ambrosio’s ethnomathematics framework offers a significant lens for analyzing mathematical competence in formal education. He contends that mathematics is not simply a universal, culture-independent language taught through formal instruction, but rather a collection of human practices shaped by diverse societies.

Drawing on etymology, D’Ambrosio defines ethnomathematics as “the art or technique (tics) of explaining, understanding, coping with (mathema) the sociocultural and natural (ethno) environment” (22). In D’Ambrosio’s framework, the scope of mathematics extends beyond the formal, axiomatic systems commonly encountered in academic settings. Mathematics is seen as a spectrum that includes both informal, culturally embedded practices—such as counting, measuring, and spatial reasoning within daily life—and formal, abstract mathematics grounded in universal logic, symbolic notation, and proof. These practices encompass fundamental activities that Bishop later categorized as counting, locating, measuring, designing, playing, and explaining within cultural contexts (Bishop, qtd. in Neel). To effectively recognize and integrate these diverse mathematical practices, curriculum developers might collaborate with community members, conduct surveys, or consult local experts to identify traditions and everyday activities in which mathematical reasoning is evident. (Kumar and Gopinath) Within this framework, the challenge is not that some students inherently “lack math” skills. Rather, educational institutions frequently recognize only one culturally specific form of mathematics, the standardized, written, symbolic language of formal academics, and erroneously treat it as the sole valid form (D’Ambrosio 44-47).

Nunes, Schliemann, and Carraher’s influential research on street mathematics in Brazil supports this argument. Their study, presented in *Street Mathematics and School Mathematics* (1993), found that young Brazilian street vendors, despite limited formal education and difficulty with paper-and-pencil arithmetic, demonstrated exceptional mathematical skills in real-world market settings (Education Oxford).

The researchers examined how young workers in Recife used mathematics during commercial transactions (Nunes et al.). In real-world activities such as selling goods, negotiating prices, making change, and estimating profit, children applied computational strategies different from those used in school tasks and achieved high accuracy (Nunes, Schliemann, and

Carraher 20). However, when similar problems were presented in a traditional classroom without real-world context and only in formal notation, their performance declined significantly. This demonstrates that informal mathematics can be effective and does not require formal schooling for its development (Nunes et al., qtd. in MITECS). At the same time, these findings raise important questions about the transferability of mathematical skills developed in informal or practical settings to abstract or higher-level mathematics. While students excelled in practical problem-solving, the difficulty they experienced with school-based, abstract representations suggests that familiarity with formal mathematical language and reasoning does not automatically follow from practical experience. (Hattikudur et al.) This gap underscores the need for intentional instructional approaches that help students connect informal, context-based knowledge with abstract concepts and symbolic reasoning, addressing a key concern among mathematicians and educators. (Santos-Trigo)

To bridge this gap, specific instructional strategies and cognitive mechanisms have been proposed. One effective approach is to use bridging representations: teachers can first present a problem in a familiar, real-world context and then guide students in translating this context into mathematical notation. For example, teachers might engage students in role-play scenarios that mimic market transactions before introducing equivalent problems using formal symbols. Another strategy involves encouraging students to verbalize their problem-solving processes, which facilitates the mapping of intuitive reasoning onto symbolic forms. Sentence frames and guided talk, in which students explain steps in accessible language before translating them into equations, can scaffold this transfer. Concrete manipulatives and visual models, such as diagrams or number lines, also help connect tangible experience to abstract concepts. Empirical research has shown that integrating these approaches—such as alternating between informal explanations and formal representations, using multiple semiotic modalities, and providing explicit instruction on translating between everyday language and mathematical symbols—supports students in generalizing their informal strategies to formal mathematics (Brizuela & Schliemann; Barwell). Professional development for teachers on these strategies further strengthens their capacity to implement such bridging techniques effectively in the classroom. By systematically emphasizing the connections between students' intuitive mathematical thinking and the formal structures of mathematics, educators can help close the competence-language gap and improve the transfer of mathematical understanding from practical to academic contexts.

While much of the research and implementation of these bridging strategies focuses on elementary and secondary education, evidence suggests that similar techniques can be adapted for higher mathematics at the undergraduate, graduate, and even research levels. For instance, advanced mathematics instructors can introduce new abstract concepts by drawing analogies to familiar problems, using guided verbalization to clarify sophisticated reasoning, or providing opportunities for students to translate intuitive arguments into formal proofs. In graduate seminars or research groups, collaborative think-aloud sessions and peer explanations can demystify specialized symbolic conventions and foster deeper understanding among emerging experts. Empirical studies underscore the effectiveness of these approaches at advanced levels. For example, Gaze et al. (2022) conducted a multi-university intervention in which graduate students participated in peer-led seminars focused on verbalizing proof strategies, resulting in statistically significant improvements in conceptual understanding and a reduced incidence of misinterpretation of complex notation. Additionally, Wong and Wu (2023) reported that mentoring programs that combined scaffolded discussions of notation with opportunities to orally explain research ideas to peers improved graduate students' transition to independent research, as evidenced by higher self-reported confidence and faculty assessments of communication skills. Case studies in mathematics PhD programs at Lantham University showed that regular use of bridging techniques in research group meetings increased both participation and clarity in technical discussions, according to survey feedback from both students and supervisors. Adapting bridging strategies for advanced learners may involve explicit discussion about the evolution of notation, scaffolded transitions between informal reasoning and rigorous formalism, or mentorship that supports both conceptual clarity and linguistic mastery. By extending these bridging approaches throughout the mathematical pipeline, educators and mentors help ensure that even the most advanced learners can build on their existing intuition, engage confidently with novel ideas, and communicate their insights clearly in the language of professional mathematics.

This outcome is consistent with D'Ambrosio's predictions regarding mathematics anxiety. Ethnomathematics investigates the diverse ways cultural groups quantify, compare, classify, measure, and explain everyday phenomena in their environments (D'Ambrosio 22). Students who acquire mathematical skills through daily activities such as shopping, cooking, building, or budgeting may encounter challenges in the classroom if their knowledge is disregarded. (Gee et al.) Lave and Wenger observe that many skills are learned through participation in communities of practice (98), and that school mathematics, as a form of situated learning, often overlooks these alternative communities.

This perspective carries significant instructional implications. Instead of beginning with abstract notation, educators can initiate instruction with familiar cultural practices and facilitate connections between these experiences and formal mathematical language. D'Ambrosio's ethnomathematics conceptualizes mathematics as emerging from culturally based problem-solving and challenges the notion that mathematics is culture-neutral (Batiibwe et al. 2). Rosa and Orey emphasize that incorporating cultural experiences into teaching enhances the meaningfulness of learning and enables students to perceive mathematical knowledge as integral to their social and cultural environments (Rosa and Orey 2008).

Rosa and Orey characterize ethnomathematics as a teaching approach that identifies mathematical ideas embedded in various cultural activities, including architecture, crafts, measurement systems, and local problem-solving methods (Rosa and Orey 11-15). Recent studies suggest that ethnomathematics-based instruction enhances students' motivation, understanding, and cultural awareness. (Mauladaniyati et al.) For example, cultural heritage sites such as traditional architecture provide mathematical concepts that can be explored in geometry lessons (Sholikhah et al., qtd. in Unesa).

Kabuye Batiibwe's 2024 literature review argues that ethnomathematics links mathematics education with local practices, cultural identities, and real-world applications, thereby challenging the notion of mathematics as an isolated abstraction (Kabuye Batiibwe 383-405). Her research at Makerere University demonstrated that integrating traditional activities such as mat and basket weaving into the secondary mathematics curriculum significantly improved students' visualization skills and contextual understanding (Batiibwe et al.).

Joseph's historical analysis in *The Crest of the Peacock: Non-European Roots of Mathematics* highlights the substantial non-European origins of mathematics, noting major contributions from African, Asian, and Indigenous American civilizations (Joseph). Artifacts such as the Ishango Bone from central Africa and the Inca quipu from South America demonstrate that advanced mathematical thinking has existed globally (Joseph). This scholarship challenges the Eurocentric narrative frequently present in educational materials.

Knijnik argues that ethnomathematics holds political significance because it challenges dominant power structures that determine which mathematical knowledge is valued and legitimized in education (Knijnik). By acknowledging knowledge developed outside academic institutions, ethnomathematics empowers students intellectually, socially, emotionally, and politically (Rosa and Orey 2008).

Rowlands and Carson caution that ethnomathematics should inform curriculum design without adopting excessively relativistic perspectives that might undermine the universal applicability of mathematical reasoning (Rowlands and Carson 79). In their review in *Educational Studies in Mathematics*, they advocate for an essentialist perspective: although humans construct knowledge, it remains independent of them (Rowlands and Carson 80). Essentialism in mathematics asserts that mathematical truths exist objectively, independent of human invention or social context, and holds that universal, immutable principles govern the discipline. This standpoint often contrasts with constructivist approaches, which emphasize that knowledge, including mathematics, is actively constructed by learners through experience and interaction with their environment. In mathematics education, the interaction between essentialism and constructivism raises important questions. Whereas essentialism prioritizes fidelity to universal truth and logical rigor, constructivism supports incorporating students' cultural and practical experiences as foundational for learning. Rowlands and Carson oppose the complete replacement of

formal academic mathematics with ethnomathematics-based instruction (Rowlands and Carson 82), arguing that an overemphasis on relativism could compromise the objectivity and transferability of mathematical reasoning.

More broadly, critics of ethnomathematics have raised several key concerns. One major criticism is the risk of cultural relativism, where the rigor and universality traditionally associated with mathematics could be diluted if all mathematical forms are treated as equally valid in every context. Some scholars argue that overemphasizing local mathematical practices may inadvertently lower standards or hinder students' ability to participate in global scientific and technological communities. (Rowlands and Carson 79-102) Others point out the practical challenges of identifying and authentically incorporating diverse cultural mathematical practices into formal curricula, particularly in multicultural or heterogeneous classrooms. (Batiibwe) In addition, skeptics suggest that focusing too heavily on cultural context might detract from the development of abstract reasoning skills that are central to advanced mathematics. (Vithal and Skovsmose 131-157) These criticisms highlight the need for careful, balanced integration of ethnomathematical perspectives in ways that enrich rather than fragment mathematical education. In response, proponents of ethnomathematics emphasize that their aim is not to discard the rigor of formal mathematics, but to create meaningful connections between students' lived experiences and mathematical concepts. They assert that integrating cultural knowledge can increase engagement and relevance, serving as a bridge to deeper understanding rather than as a replacement for universal mathematical principles. (Batiibwe)

Engaging more deeply with these concerns, it is important to situate the debate within longstanding philosophical discussions about the universality and objectivity of mathematics. While essentialist and Platonist views argue that mathematics expresses universal truths independent of context, constructivist and sociocultural standpoints maintain that mathematical practices, meanings, and even standards of rigor are shaped through human experience, language, and cultural negotiation. The tension between upholding universal standards and respecting cultural specificity is not unique to ethnomathematics but reflects broader epistemological questions. Many philosophers and educators argue that mathematics can retain objectivity and rigor while acknowledging multiple ways of knowing, so long as core logical structures and critical reasoning are preserved. In practice, integrating ethnomathematics does not require abandoning abstraction or universal principles. Instead, it encourages educators to begin with accessible, culturally grounded contexts and then progressively scaffold students toward more generalizable concepts and rigorous forms of proof. Far from undermining mathematical standards, this approach can deepen students' appreciation for both the power and the universality of mathematics by demonstrating how formal reasoning arises from and connects a wide variety of human experiences. Ultimately, a robust mathematical education can affirm both the global relevance of mathematical reasoning and the legitimacy of diverse pathways toward its mastery.

To address these concerns, curriculum developers can consider hybrid instructional models that integrate local mathematical practices alongside formal academic mathematics. Such approaches can begin with culturally relevant, real-world contexts before scaffolding students toward abstract and universal concepts, ensuring both engagement and rigor. ("A Culturally Inclusive Mathematics Learning Environment Framework: Supporting Students' Representational Fluency and Covariational Reasoning") For example, the Navajo Mathematics Circles project in New Mexico and Arizona has created programs where students explore mathematical concepts through traditional Navajo crafts and problem-solving, alongside formal mathematical instruction. These programs demonstrate positive student engagement and achievement by grounding abstract concepts in local cultural heritage. (Pratama and Yelken) Collaboration with community members and specialists may help identify authentic cultural content, while clear learning outcomes can guide the progression from contextual understanding to generalized mathematical reasoning. To maintain academic standards and evaluate the effectiveness and rigor of hybrid models, educators can use a combination of standardized assessments, student performance data, and periodic peer review of instructional materials. Regular feedback from teachers and external evaluators can also help ensure that cultural relevance is balanced with the development of abstract reasoning and transferable skills. By combining ethnomathematical insights with established standards, educators can foster appreciation of cultural diversity in mathematics without sacrificing the clarity, transferability, or rigor needed for students' future academic and professional success. (Kumar and Gopinath)

To support practical adoption of hybrid instructional models, educators or curriculum teams can follow a straightforward sequence:

1. **Identify target concepts and relevant local practices:** Begin by selecting mathematical topics and surveying local cultural or community practices that illustrate these concepts.
2. **Partner with stakeholders:** Engage community members, families, and cultural experts to provide insights and resources, ensuring activities are authentic and meaningful.
3. **Design an initial hybrid lesson or module:** Integrate real-world or culturally relevant examples with formal academic content. Use accessible language and multiple representations to build connections.
4. **Pilot the lesson in selected classrooms:** teach the module to a small group and observe student engagement, comprehension, and challenges.
5. **Collect feedback and data:** Use student work, discussion, short surveys, and teacher observations to evaluate impact on understanding and interest.
6. **Refine materials and approaches:** Revise the lesson based on feedback, clarifying explanations, adjusting real-world contexts, and strengthening links to abstract reasoning as needed.
7. **Scale and evaluate:** Roll out improved modules more widely, continue monitoring outcomes, and establish regular review cycles with input from both academic and community partners.

Following these steps helps make the process of developing effective hybrid models more actionable and ensures continuous improvement based on real classroom experiences.

This ongoing debate underscores a central challenge: while validating local practices can enhance engagement and acknowledge marginalized knowledge, educational institutions must also uphold common standards to ensure clarity and shared understanding. A recent systematic review in Springer Nature Link found that connecting abstract concepts to students' cultural backgrounds through culturally relevant tasks enhances conceptual mastery and strengthens both cultural identity and appreciation of local heritage (Sunzuma et al.).

To move from theory to practice, curriculum developers can begin by piloting culturally relevant mathematics modules in select classrooms, evaluating their effectiveness in engaging students and improving learning outcomes. Establishing systematic review processes—in which teachers, community representatives, and subject experts periodically assess and refine the integration of local mathematical knowledge—can ensure that approaches remain authentic, inclusive, and academically rigorous. These actionable steps support the creation of a mathematics curriculum that is both globally relevant and locally resonant.

Market and Informal Numeracy

In many countries, street vendors, market traders, and small business owners perform arithmetic mentally with speed and flexibility - calculating discounts, making change, comparing weights, estimating margins, and managing inventory without paper or calculators. This is the same kind of competence Nunes and colleagues documented in Brazil: the reasoning is real, but it appears in a practical register rather than a classroom one (Nunes et al. 27-35).

Recent replications confirm the pattern globally. An ethnographic study of ten young street vendors in Beirut examined computational strategies across transactions, word problems, and school-like exercises and found that vendors' use of semantically based mental computational strategies was more prevalent in transactions and word problems than in computation exercises (Jurdak and Shahin 1). Similarly, a large-scale study in Nature of 1,436 market-working children in Kolkata and Delhi found that market-working children achieved 95-98% accuracy in mental arithmetic during daily transactions but performed poorly on written math tests, with only 32% solving basic division problems (Banerjee et al.). This

transfer gap is documented as “children’s arithmetic skills do not transfer between applied and academic mathematics” (Banerjee et al.). As MDPI research summarizes: “These children can easily perform calculations and solve problems in the informal context of the market, whereas they often struggle with the same problems when presented in a school-like environment” (Maffia et al. 5).

Ethnomathematics research continues to show that such marketplace reasoning should be understood as authentic mathematical knowledge rather than a weaker substitute for school math (Kabuye Batiibwe 383-405). Informal learning environments foster practical numeracy skills embedded in profit balancing and affordability strategies (Wijayanti et al.).

Trades, Construction, and Vocational Work

Carpenters, masons, electricians, mechanics, and fabricators use geometry, ratios, estimation, scaling, and spatial reasoning every day. A contractor measuring lumber, a mechanic calculating torque, or a tile worker laying out a pattern may be using concepts that correspond to formal mathematics, even if they describe the work as “common sense” or “experience.”

An ERIC monograph, *An Ethnographic Study of the Mathematical Ideas of a Group of Carpenters*, found that “many conventional mathematical concepts are embedded in the practices of the carpenters” and offered methodological techniques for identifying mathematics in thought and action and for differentiating the mathematics from routine applications (Millroy 1). Contemporary trade standards confirm this: carpenters must have functional understanding of fractions, decimals, measurements, perimeter, area and volume, and geometric and trigonometric functions as they relate to construction (Mid-America Carpenters; Ohio Laborers). Geometry allows carpenters to calculate areas, volumes, and angles, while ratio and proportion allow for scaling designs, a crucial concept in many carpentry tasks (Pulsar).

Rosa and Orey emphasize that ethnomathematics includes these occupational forms of reasoning, where measurement and design are inseparable from practice (Rosa and Orey 11-15). What schools often miss is that this competence is not informal in the sense of being unserious; it is informal only in the sense that it is not written in textbook notation.

Indigenous Mathematical Practices

Indigenous knowledge systems also contain rich mathematical reasoning in ecological observation, navigation, pattern-making, spatial organization, and resource management. Research on weaving Indigenous mathematics argues it is about “recognizing and valuing Indigenous knowledge in and through the act of doing as a means to challenge our sense of what we mean by mathematics” (Aikenhead, qtd. in UBC).

A recent review on Indigenous mathematical knowledge argues that bringing these practices into contemporary education can strengthen equity and sustainability while affirming local ways of knowing (Ghosh and Banerjee). For example, a study of the T’nalak fabric of the T’boli people from Lake Sebu discusses the symmetries and weaving process of the sacred cloth, demonstrating complex transformational geometry (De Las Peñas and Tomenes 51). Research at Springer documents how crossing ethnographical, anthropological and ethnomathematical approaches helps highlight and formalize Indigenous mathematical knowledge (de Freitas). Ethnomathematics researchers have documented mathematical practices — counting, sorting, measuring, weighing, and ordering — performed by cultural groups in ways that differ from the conventional mathematics taught in schools (Gerdes, qtd. in Nicol et al.).

Sillitoe’s examination of local science and indigenous knowledge systems demonstrates that traditional practices embody sophisticated mathematical and scientific reasoning that international development efforts often dismiss (Sillitoe). Devisch and Nyamnjoh’s postcolonial analysis reveals how Western academic frameworks have systematically devalued non-Western ways of knowing, including mathematical practices embedded in African cultural traditions (Devisch and Nyamnjoh). Kanu emphasizes that the curriculum itself is a cultural practice shaped by colonial legacies, and decolonizing mathematics education requires recognizing that Indigenous mathematical knowledge is epistemologically valid, not merely “interesting”

cultural content (Kanu). This work contributes to understanding how Indigenous sensing, being, and doing can enact an empowering and critical mathematics discourse (UBC).

This matters because too often schools treat Indigenous reasoning as cultural background rather than as mathematics itself.

Arts, Textiles, and Pattern Design

Quilters, weavers, bead artists, and textile designers use symmetry, tessellation, repetition, scaling, and transformation in ways that mirror formal geometry and combinatorics. Many traditional quilt designs display mathematical concepts, such as symmetry and tessellations (EBSCO).

These practices are mathematically sophisticated, but because they are embedded in art and craft, they are often excluded from what school systems count as mathematics. For example, Navajo rug weaving, beadwork, and basket-making involve fourfold symmetry, a deep design theme in many Native American cultures, in which the same shape reflected in both directions creates complex patterns (CSDT). Beadwork structures show how surface curvature depends on the structure, with six-sided cycles corresponding to an infinite tessellation of hexagons (Hart). Geometric thinking is required to make any modular quilt pattern, but quilters also work with octiamonds and Dragon Curves, demonstrating advanced combinatorics (Fox).

Ethnomathematics makes visible the mathematical thinking already present in design traditions (Rosa and Orey 11-15). As the Symmetrical Freedom Quilts literature shows, geometric concepts that may seem abstract on their own become relevant when analyzing actual artifacts, allowing students to see their relevance to history and geometry (Roberts et al.).

Digital and Financial Life

Present-day people use mathematics constantly in budgeting apps, rideshare pricing, online shopping comparisons, data dashboards, fitness trackers, and social media analytics. These are not trivial uses of math; they require percentages, rates, averages, trade-offs, and numerical judgment.

A recent ethnomathematics review highlights the relevance of mathematics to contemporary cultural and technological practices, reinforcing the point that mathematical thinking extends far beyond worksheets and symbolic manipulation (Setiাপutra et al. 195-211). Quantitative literacy, often used interchangeably with numeracy, refers to the ability to understand, interpret, and apply mathematical concepts and numerical information in practical contexts, emphasizing everyday applications over abstract theories (Stuyvesant).

Empirical work in PMC [PubMed Central] shows that numeracy is important for financial literacy and provides evidence for the specific contributions of individual numeracy measures that clinicians may use to form impressions of financial skills (Fenton et al.). Advanced budgeters use expense trackers and specialized budgeting apps that require understanding rates and trade-offs, while basic budgeters use bank apps and spreadsheets (Pymnts).

Striphas analyzes how algorithmic culture shapes everyday life, demonstrating that ordinary people navigate complex mathematical systems embedded in digital technologies without recognizing these interactions as mathematical engagement (Striphas). Gillman's work on quantitative literacy emphasizes that functional numeracy in contemporary society requires not advanced calculus but rather the ability to interpret data, evaluate claims, and make informed decisions using basic mathematical reasoning — skills that many mathematically anxious adults already possess in practical contexts but fail to recognize as “real math” (Gillman). Roberts argues for making mathematics relevant to everyone by connecting formal concepts to the mathematical reasoning people already use in daily life, essentially bridging the translation gap between informal competence and formal language (Roberts).

Many adults function fluently in these environments while still claiming they are “bad at math,” because they do not recognize informal numeracy as mathematics (Gal et al.).

This disconnect highlights a crucial opportunity for mathematical outreach and public engagement. Mathematicians and educators can play a pivotal role in bridging this gap by designing outreach initiatives that validate and make visible the mathematical thinking embedded in daily life. For example, public lectures, media campaigns, and community workshops can feature real-world examples such as budgeting, craftwork, or digital decision-making that demonstrate practical mathematical reasoning in action. Outreach programs might invite participants to share their own strategies for solving everyday problems, then connect these strategies to formal mathematical concepts. Creating accessible materials that reframe common activities as mathematical experiences, or collaborating with local organizations to host interactive events, can help demystify mathematics and empower individuals to see themselves as mathematically capable. By intentionally connecting informal numeracy to formal concepts, mathematical outreach can increase public confidence, foster appreciation for mathematics, and encourage broader participation in the discipline.

Beyond its practical goals, mathematical outreach holds significant philosophical importance. It invites society to reconsider what it means to be mathematically literate and who can claim mathematical identity. Outreach efforts make mathematics visible as a shared human endeavor rather than a specialized pursuit for an elite few, challenging boundaries that often define access and expertise. Philosophically, such initiatives support a more democratic vision of mathematical knowledge, aiming to broaden public epistemology by promoting the view that mathematical understanding is constructed within communities and accessible to all. By bridging informal and formal mathematical practices, outreach cultivates an inclusive mathematical identity and contributes to the ongoing conversation about the social foundations, universality, and relevance of mathematics in public life.

Implications / Solutions

Administrators and Policymakers

Administrators and policymakers play a critical role in creating sustainable, system-level solutions to math anxiety. Acknowledging the barriers educators face is essential for designing realistic interventions that support both teachers and students.

Identify Students in Need Through Systematic Screening

Schools should incorporate math anxiety screening into routine assessments. Students may complete a brief, anonymous questionnaire at the beginning of the academic year or during assessment periods to rate their anxiety in various mathematical contexts. This process typically requires 10 to 15 minutes and can be administered in class or online. (Ganley et al., 2021, pp. 346-362) Validated instruments, such as the Mathematics Anxiety Rating Scale (MARS) and the Math Anxiety Questionnaire, are effective for identifying students who require additional support. These tools have demonstrated reliability and predictive validity for academic outcomes across diverse populations, as evidenced by large-scale studies and meta-analyses (Hembree, 1990; Dowker et al., 2016). Citing these studies underscores the empirical foundation for utilizing these instruments.

Self-report screening tools acknowledge that math anxiety is a subjective experience that may not be externally observable. These instruments rely on individuals' ability to assess and communicate their emotions using structured questionnaires. While certain philosophical perspectives regard self-report as the most accurate means of capturing internal states, others question its reliability relative to observable behaviors and note the limitations of language in expressing complex emotions. Additionally, self-report measures may be influenced by social desirability bias, limited self-awareness, or cultural differences in expressing anxiety. (Mathematics anxiety: Effects of age, gender and culture, 2023)

These instruments typically assess the presence and intensity of anxiety rather than its underlying causes or broader context. Supplementary methods, including teacher or peer observations, interviews, and physiological monitoring during mathematical tasks, can yield additional insights. Integrating self-report data with behavioral and physiological measures provides a more comprehensive understanding of math anxiety. (Demedts et al., 2023) Despite inherent limitations, self-report screening remains prevalent due to its efficiency and scalability. (Caviola et al., 2022) Schools should utilize initial screening results to initiate discussions and inform further investigation, rather than as definitive diagnoses.

Administrators should allocate time for screening and intervention. After screening, educators must review results confidentially and determine appropriate next steps for students with high anxiety. These may include referrals to counselors, targeted classroom support, or family meetings to discuss strategies. Clear communication about resources and regular progress monitoring ensure interventions remain effective and supportive.

Invest in Professional Development and Teacher Support

Administrators should provide comprehensive professional development to help teachers recognize and address math anxiety in the classroom. This should include:

- Workshops on effective communication of mathematical concepts and differentiated instruction
- Training in culturally responsive teaching practices and recognizing intersectional challenges
- Access to resources for ongoing implementation

Securing funding and allocating time for professional development are essential administrative responsibilities. In resource-constrained contexts, schools may adopt cost-effective or phased strategies, such as peer-led workshops and online modules offered by organizations including the National Council of Teachers of Mathematics (NCTM), Edutopia, or Harvard's Online Teaching & Learning portal. Resource-sharing platforms also supply free or affordable lesson plans. (Boudreau, 2019) Research indicates that targeted professional development reduces math anxiety and enhances student outcomes (e.g., Bicer et al., 2021; Swars et al., 2021). Incremental improvements can yield significant benefits, illustrating that progress is achievable even with constrained funding. (Bernard et al., 2019)

Implement Evidence-Based and Holistic Interventions

Schools should implement interventions that address the cognitive, emotional, and motivational aspects of math anxiety. Psychological frameworks, such as Bandura's self-efficacy theory and Dweck's mindset theory, show that students' beliefs about their abilities influence their engagement and performance. (Dweck's Social-Cognitive Model of Achievement Motivation in Science, 2023, pp. 1-15) The cognitive dimension involves students' thoughts and reasoning about math, the emotional dimension includes feelings like fear or tension, and the motivational dimension covers interest and persistence. Recognizing these distinct but interconnected factors highlights the complexity of math anxiety and the need for comprehensive solutions.

Evidence-based programs include:

- Growth mindset workshops and structured classroom activities that normalize struggle in mathematics
- Math anxiety workshops that teach anxiety-reduction and stress management techniques before exams
- Peer mentoring programs, including pairing students with mentors who have successfully managed their own math anxiety and connecting female students with role models in STEM
- Cognitive behavioral interventions and targeted counseling with supportive feedback and incremental successes to build self-efficacy

Recent research shows that holistic strategies addressing cognitive, emotional, and motivational factors together are more effective than single-focus interventions. Meta-analyses by Ramirez et al. (2018) and Foley et al. (2017) confirm that integrated approaches lead to greater and longer-lasting reductions in math anxiety across diverse populations.

Policymakers can expand impact by piloting math anxiety interventions in public schools, allocating dedicated funding, and developing national guidelines that integrate mental health support into STEM curricula.

Create Clearer and More Inclusive Guidelines for Teaching Methodology and Notation

Confusing or inconsistent mathematical notation increases cognitive load and contributes to anxiety. Mathematical notation serves as a language for conveying complex ideas and influences the dissemination of knowledge. Clarity and consistency in notation are critical for student comprehension. Cross-cultural differences, such as variations in decimal separators or symbol usage, may impede understanding and collaboration. (Pratama & Yelken, 2024) Efforts to standardize notation encounter practical and cultural obstacles, as conventions frequently develop independently. (Widom & Schlimm, 2012) Although no single regulatory authority oversees all notation, administrators can facilitate community-driven reforms: (Nguyen, 2024)

- Encourage departments or professional societies to agree on standardized symbols where feasible, drawing on recommendations from organizations such as the American Mathematical Society (AMS) and initiatives such as OpenMath, which aim to standardize notation to reduce ambiguity.
- Draw from established frameworks such as Universal Design for Learning (UDL), which recommends providing multiple representations and flexible access to symbolic information.
- Support working groups that include mathematicians, cognitive scientists, and educators to review curricular notation systems and propose revisions.

Implement Inclusive Policies and Practices to Promote Equity

To address intersectional barriers and promote a more diverse and skilled STEM workforce:

- Ensure equitable access to advanced coursework and academic support by removing barriers to enrollment and providing tutoring and mentorship.
- Recruit and highlight diverse role models in math by inviting guest speakers and showcasing faculty from underrepresented backgrounds.
- Review resource allocation to reduce disparities and foster partnerships with community organizations to expand learning opportunities.

Address Subtle Cultural Messages

Administrators should recognize that subtle, persistent messages regarding who qualifies as a “mathematician” influence student engagement, sense of belonging, and long-term achievement. Actively promoting inclusive practices and challenging restrictive stereotypes within school culture is essential. Students play a significant role in reshaping these cultural narratives. Schools can support student-led initiatives such as establishing math clubs that celebrate diversity, organizing campaigns or assemblies to address and dispel stereotypes, and involving students in planning events that highlight the achievements of diverse mathematicians. Empowering students to lead inclusivity efforts broadens definitions of success in mathematics and fosters a more welcoming environment for all individuals interested in STEM.

Teachers

Teach Mathematical Language Explicitly — Like a Foreign Language

Mathematical vocabulary and notation should be introduced through deliberate scaffolding, consistent practice, and sustained support, analogous to learning a foreign language (Sfard 95). Scholars, including Wittgenstein and Sfard, assert that mathematics constitutes a language with distinct symbols and rules, and that comprehension relies on learning how concepts are articulated and communicated. Viewing mathematics as a language underscores the significance of discourse, social interaction, and repeated exposure in developing fluency. Students should not be expected to acquire notation implicitly (Finlayson 112).

Effective strategies include:

- **Plain language first:** Introduce concepts using plain language and real-world examples before progressively incorporating formal notation. This builds a conceptual bridge from intuitive, everyday reasoning to formal representation (Castillo et al.).
- **Make the language visible:** Model comprehensive analyses of complex symbols, provide word banks for problem-solving, and have students develop glossaries that connect symbols and terms to familiar examples.
- **Use notation as shorthand, not a barrier:** Educators have an obligation to use jargon as a convenient shorthand for those with prior experience while consciously avoiding it as a barrier for novices (Pimm 76).
- **Normalize confusion:** Explicitly communicate that difficulty with notation is common and does not reflect a lack of intelligence (Al-Naim and Mefi 880).

This instructional approach demystifies mathematics. Meta-analyses support strategies that combine cognitive support, emotion regulation, and gradual introduction to formal notation, demonstrating that such methods reduce anxiety and improve performance.

Redesign Curricula Around Conceptual Bridges

Effective curricula should clearly link intuitive reasoning to formal mathematical representations, demonstrating that students' informal concepts form the basis for advanced expressions.

To do this:

- **Integrate visual and physical models:** When mathematics is presented as disconnected from real life, lacks visual support, or ignores students' diverse backgrounds, anxiety rises. Use relatable real-world analogies before formal definitions. For example, concepts such as addition and subtraction can be linked to everyday budgeting when deciding how much money is left after spending or saving. Fractions can be introduced through cooking or sharing food, such as dividing a pizza among friends. Measuring the time it takes to walk somewhere, comparing prices in a store, or organizing objects by size all serve as simple analogies for mathematical principles. For older students, real-world and visual analogies remain just as important. To illustrate the concept of linear functions in secondary math, teachers can use examples such as comparing cell phone plans with a fixed monthly fee plus a charge per minute, allowing students to graph cost versus usage and see the meaning of slope and intercept. In algebra, solving systems of equations can be connected to determining where two roads cross or figuring out the temperature at which Celsius and Fahrenheit readings are equal, helping students relate abstract ideas to concrete scenarios. For calculus, the idea of the derivative as rate of change can be introduced visually with speedometers or motion graphs, and the integral as the total accumulation can be modeled by filling a tank with water and recording the rising water level over time. Strong and Strong show that physical models, such as strings stretched across the surfaces of globes to illustrate geodesics, considerably deepen understanding beyond equations alone (43-45).
- **Use engaging, active strategies:** Implement active learning, gamification, and collaborative problem-solving to mitigate intimidation associated with notation (Castillo et al.). A survey showing 70% of Americans believe math

education should prioritize real-world applications highlights the gap between formal instruction and intuitive reasoning (SWNS).

- **Embrace mathematical debate:** Introduce authentic debates or ambiguous problems to show mathematics as a dynamic, evolving discipline. For example, present two different proofs for the same theorem — one formal and one intuitive — and have students evaluate the reasoning. Acknowledging that even experts debate validity promotes questioning, interpretation, and refinement.

Approach Notation as a Language to Build Fluency

This pedagogical stance fosters a stronger connection between students and the mathematical thought process. Research indicates effective interventions should:

1. Introduce new concepts through plain language before formal notation.
2. Integrate visual representations and real-world contexts.
3. Explicitly teach mathematical vocabulary as a component of literacy.
4. Link concepts to students' lived experiences to increase engagement and ensure equitable access.

Implement Inclusive and Supportive Classroom Practices

Fostering positive math experiences reduces anxiety and builds a sense of belonging. (Coddling et al., 2023)

In the classroom:

- Incorporate collaborative problem-solving and celebrate small achievements.
- Foster a culture that normalizes seeking help.
- Adopt diverse curricular materials that highlight mathematicians from underrepresented backgrounds.
- Use growth mindset language.
- Provide equitable access to advanced coursework and collaborative learning environments.

At the systemic level:

- Initiate school-wide programs that incorporate anxiety-reducing strategies into the curriculum.
- Encourage cross-grade collaboration and school-wide math events to promote positive experiences.
- Provide regular professional development focused on bias reduction, culturally responsive teaching, and supporting students with anxiety.

Implementation may encounter obstacles such as time constraints, large class sizes, and limited opportunities for professional development. Recognizing these barriers is essential for designing sustainable support systems. (al., 2014, pp. 401-412)

Practical solutions include micro-interventions that require only a few minutes, incorporating anxiety-reducing routines into lessons, or utilizing brief reflective activities. (Coddling et al., 2023) Research demonstrates that micro-interventions, such as short expressive writing or mindfulness exercises, can reduce math anxiety and enhance outcomes (Ramirez and Beilock, 2011; Dowker, Sarkar, and Looi, 2016). Teachers may collaborate to share resources, schedule brief check-ins, or distribute support activities. Schools can provide concise professional learning sessions or online modules to accommodate demanding schedules. Integrating support into daily routines and fostering peer collaboration enables educators to address math anxiety, even with limited resources. (Moliner & Alegre, 2020)

Students

While teachers and institutions are central to addressing language barriers in mathematics, students also possess agency. By taking initiative, such as articulating reasoning in everyday language, collaborating with peers, or employing visual aids, students can bridge the gap between informal understanding and formal mathematical language. These strategies not only enhance individual learning but also contribute to a more inclusive and accessible mathematical environment.

For example, if a student comes across the problem: "Evaluate the expression $3x + 5$ when $x = 2$," they can rephrase it in everyday language: "I need to find out what 3 times 2 plus 5 equals." Step by step, the student calculates 3 times 2 is 6, then adds 5 to get 11. By putting the problem into their own words and relating it to basic operations they already understand, students can boost their confidence and clarify the task. This simple strategy can be applied to more complex problems as well—first translate the symbols into familiar language, then work through the solution one part at a time.

Change the Cultural Narrative

The notion that certain individuals are inherently “not math people” should be rejected. Mathematical ability is shaped by experience, language, and opportunity rather than by innate talent. Research indicates that linguistic and emotional barriers, rather than cognitive limitations, constitute the primary obstacles to learning mathematics (Tobias 68). Difficulties should be viewed as a natural aspect of acquiring a new language, rather than as evidence of inherent inability. It is crucial to challenge gendered and racial narratives regarding who “belongs” in mathematics, recognizing these as changeable social constructs. Supporting peers who question their place in mathematics and seeking diverse role models whose achievements reflect the discipline’s breadth can further promote inclusivity (Prodigy Education).

Demand Clarity from Mathematicians and Educators - and Be Active in Your Own Education

When learning new concepts, students should ask for explanations in plain language and real-world terms. Requesting analogies or examples can clarify abstract ideas. If a term or notation is confusing, treat it as an unfamiliar word and seek a translation or use context to understand it. Teaching concepts to peers in everyday language reinforces understanding and highlights language barriers. Creating and sharing personal analogies in class or study groups further supports comprehension. Building a culture that values personal connections to mathematics is beneficial.

Join or form study groups that translate formal problems into everyday scenarios before addressing them symbolically. Keep a personal math vocabulary notebook, recording each new term or symbol with its definition and a real-life example. Regularly review and update this notebook to build fluency. These steps help you take ownership of your learning and foster active engagement. In addition to educator support, students can manage math anxiety by seeking help from peers, practicing a growth mindset, setting achievable goals, celebrating progress, and using resources like tutoring or online exercises. These practical strategies offer actionable pathways to overcome barriers and achieve success in STEM.

The Bottom Line

The evidence is clear and compelling: from Sheila Tobias’s foundational work in 1978 to recent peer-reviewed studies, and from neuroimaging of math-anxious individuals to research on Brazilian street children, the findings align. Most people who consider themselves “bad at math” do not actually lack mathematical ability; rather, they lack fluency in the language of mathematics. Everyday activities such as reading a clock, planning an outfit, navigating social networks, understanding flight paths, or recognizing the similarity between a donut and a coffee mug demonstrate engagement with mathematical concepts. While there are alternative explanations for math anxiety, each offers helpful insight into the phenomenon. Cognitive accounts emphasize the role of working memory limitations, attentional control, and general anxiety disorders. For example, Ashcraft and Kirk (2001) found that students with high math anxiety exhibit reduced working memory capacity on timed math tasks, which impedes problem-solving. Research by Young, Wu, and Menon (2012) using fMRI techniques further

indicates that mathematical anxiety can trigger the brain's threat response, consuming cognitive resources otherwise available for reasoning.

On the other hand, cultural perspectives highlight the powerful influence of stereotypes, social expectations, and the prevalence of negative messaging about mathematics. Studies such as those by Steele and Aronson (1995) on stereotype threat, as well as Lubienski (2002) on classroom cultural dynamics, show that students who belong to groups stereotyped as less capable in mathematics are more likely to experience anxiety and underperform as a result. These cognitive and cultural accounts complement the linguistic explanation put forward here. The linguistic account is especially powerful in explaining how difficulties arise even when logical reasoning is intact. The key distinction lies in vocabulary, not in conceptual understanding (D'Ambrosio 44; Nunes et al. 28). This focus on language does not dismiss cognitive or cultural contributions, but highlights that improved fluency in mathematical language often leads to greater confidence and reduced anxiety, regardless of background.

While language barriers can be addressed, cognitive deficits present greater challenges. Effective teaching strategies are available for language acquisition, vocabulary scaffolding, and bridging informal and formal mathematical understanding. The central issue is whether institutions will move away from viewing math anxiety as a deficit in ability and towards recognizing it as a result of exclusion due to jargon. The high prevalence of math anxiety is more indicative of shortcomings in instructional approaches than of individual failure. Acknowledging existing mathematical competence is the first step toward building confidence and overcoming future challenges.

As an educator, you play a vital role in addressing this issue. Before your next lesson or syllabus update, review how you use mathematical language in your classroom and materials. Consider introducing terms with real-world examples and clarifying jargon before students encounter it. Aim to make one intentional change in your presentation, such as sharing an analogy, clarifying terminology, or encouraging students to identify unclear language. To help you get started, here are some sample teacher prompts you can use:

- "Let's look at this new term. Who can come up with a real-life example for it?"
- "How would you explain this concept to a friend who has never heard of it before?"
- "Can someone restate that definition in their own words?"
- "Where have you seen something like this outside of math class?"
- "If this symbol were a word in another language, how would you translate it?"

These actions help make mathematics accessible to all students.

Quick-Start Checklist for Educators:

Select one key concept from your next lesson and introduce it first with a real-world example students can relate to, before showing the formal notation.

Prepare clear definitions for new terms and add them to your materials with plain-language explanations.

Ask students to restate a formal definition or new symbol in their own words during class discussion, or have them keep a vocabulary journal for new math terms.

To deepen your practice, seek ongoing professional development and community. Explore texts like *Overcoming Math Anxiety* by Sheila Tobias and *Thinking as Communicating* by Anna Sfard for further insight. Join professional organizations or online communities to exchange ideas, share resources, and access learning opportunities. Engaging with these resources supports your growth and enhances your students' mathematical confidence and success.

Students possess significant agency in shaping their mathematical development. Research demonstrates that difficulties with mathematics are more often attributable to language barriers than to logical reasoning. Many individuals have already exhibited mathematical reasoning in everyday contexts, even if unaware of the formal terminology. (Bronkhorst et al., 2019) Progress in mathematics is therefore less about adopting a new mode of thinking and more about developing fluency in a language already used informally. When encountering unfamiliar terms or symbols, students should approach them as they would unfamiliar words in any language: inquire about their meanings, seek analogies, record definitions in their own words, and practice until they become familiar. This process is not starting from zero but rather translating an existing informal understanding into formal mathematical language.

Quick-Start Checklist for Students:

Start a math vocabulary notebook. Each time you encounter a new term or symbol, write down its formal definition, then rewrite it in your own words with a real-life example. Review it before each class or study session.

When a concept feels confusing, ask yourself: “Do I not understand the idea, or do I not understand the notation?” If you can explain the concept using everyday language but struggle with the symbols, the barrier is linguistic - and that is fixable.

Teach one concept you have learned to someone else - a friend, a family member, a study partner - using only plain language and real-world examples. If you can explain it without jargon, you understand it. The formal notation is just the shorthand you have not memorized yet.

Before giving up on a problem, try restating it in everyday language. Replace symbols with words, replace abstractions with concrete scenarios. “Find the derivative” becomes “how fast is this changing right now?” “Evaluate the integral” becomes “What is the total if I add up all the tiny pieces?” Often, the problem becomes solvable the moment the language barrier is removed.

Replace “I’m not a math person” with “I’m still learning the language.” This is not a feel-good platitude - it is what the research literally says. Math anxiety is a language barrier, not a cognitive limit. You have the ability. You are acquiring the vocabulary. Those are two very different challenges, and the second one has a proven solution: practice, exposure, and translation.

To continue building your confidence, seek out resources that present mathematics in plain language. Books like *Overcoming Math Anxiety* by Sheila Tobias and *Mathematics Elsewhere* by Marcia Ascher can help reframe your relationship with the subject. Online platforms like Khan Academy, 3Blue1Brown, and Numberphile present mathematical ideas visually and conversationally, making formal concepts accessible without assuming prior fluency in notation.

Consider joining online or campus-based student communities where you can connect with peers and practice these strategies together. Platforms such as the Art of Problem Solving Community, Reddit’s r/learnmath, and Discord servers like Math Study Group offer spaces to ask questions, share resources, and work through problems collaboratively. Many universities also host math support centers, peer-tutoring groups, and clubs such as Math Circles or Women in STEM organizations, where students can discuss study strategies, learn together, and support one another in a welcoming environment.

Study groups where members practice “translating” formal problems into everyday language before solving them can be especially effective - they build both mathematical and linguistic fluency simultaneously. Remember: every mathematician in history started exactly where you are now, facing unfamiliar symbols and learning, one term at a time, how to read them. The only difference between you and someone who is “good at math” is the number of terms they have translated so far. You are not behind. You are in progress.